Getting Started with SAS Intelligent Decisioning

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- Create and Test a DS2 Code File
- Create and Test a Decision
- Exercise Completed

Exercise Description

In this hands-on exercise, you'll learn how to create a simple decision step-by-step. You will use the SAS Intelligent Decisioning interface to create rule sets and other decision elements, arrange decision objects in a workflow, and test the completed decisions for accuracy.

Log in to SAS Viya

Open a new window in the *Google Chrome* browser and select the **SASLanding** bookmark.

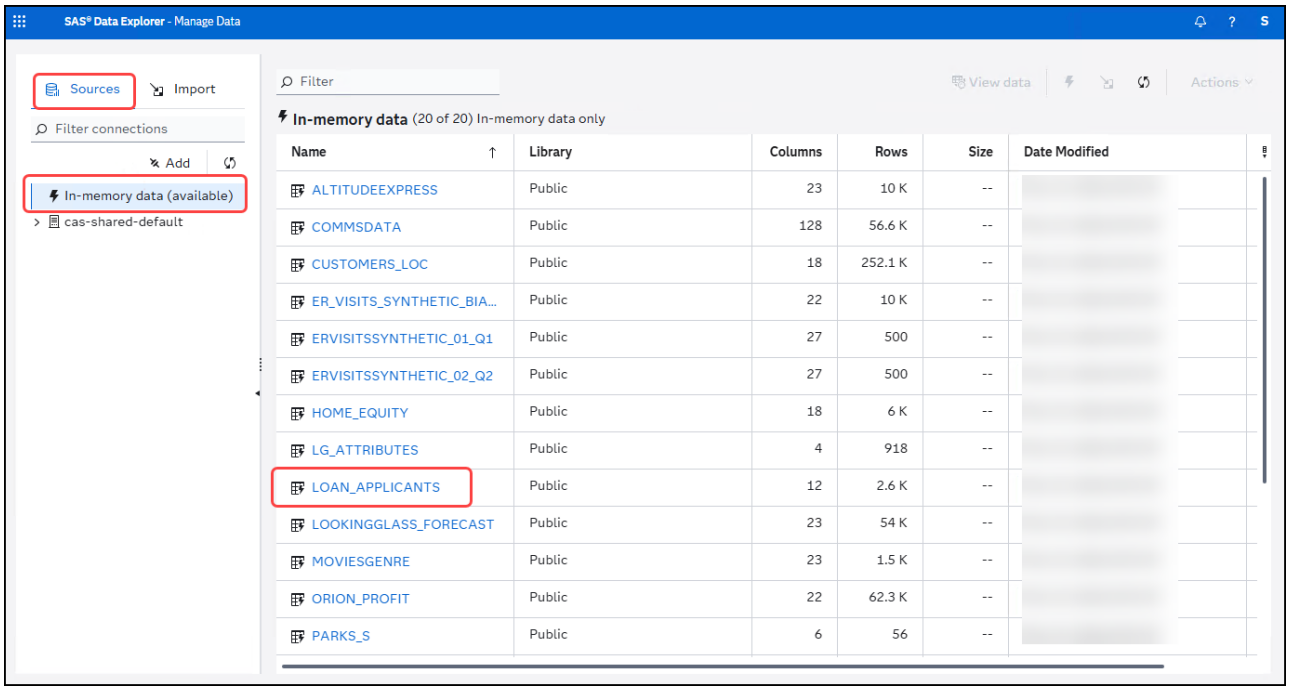
- ID: **student**
- Password: **Metadata0**

Select **No** when prompted about accepting *Admin* privileges.

View the Test Data

1. Select ➔ **Manage Data** to open *SAS Data Explorer*.

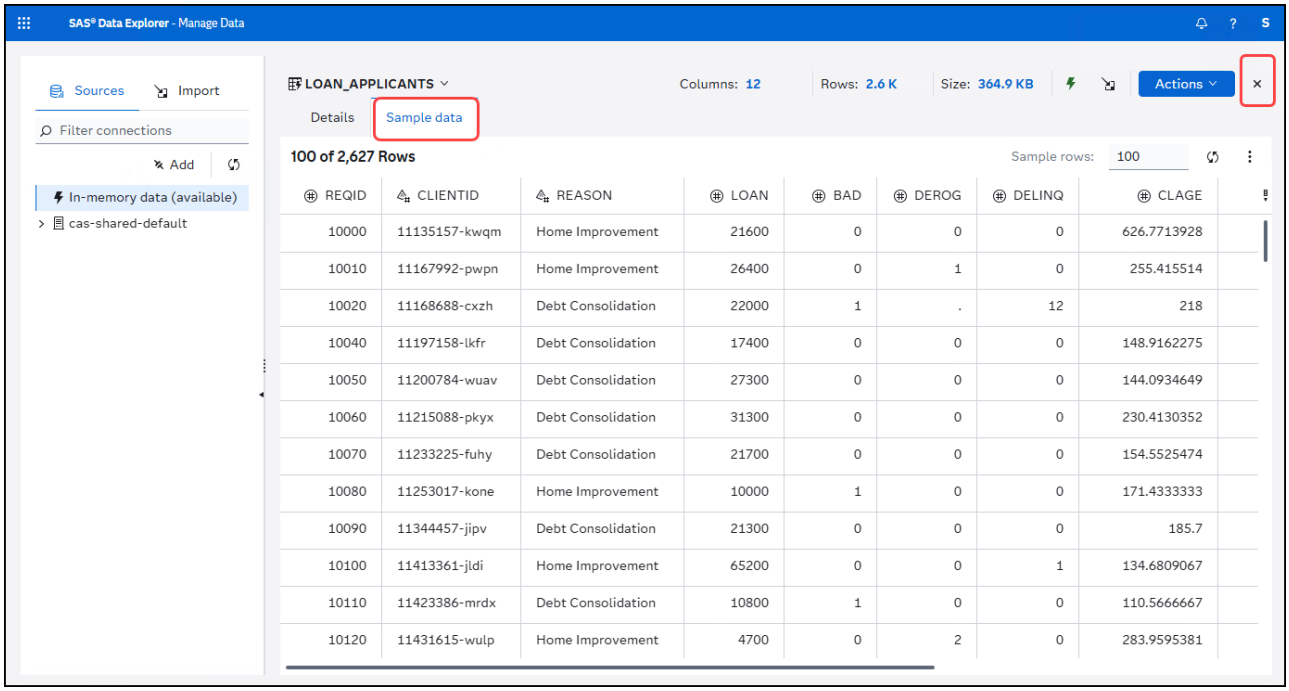
2. Select the **In-memory data (available)** source to confirm the **LOAN_APPLICANTS** table is listed there.



3. Select the **LOAN_APPLICANTS** table hyperlink to view its *Details* information.

4. Select the **Sample data** tab to view a sample of the data set.

5. Click the **x** in the right-corner of the window to close the table.

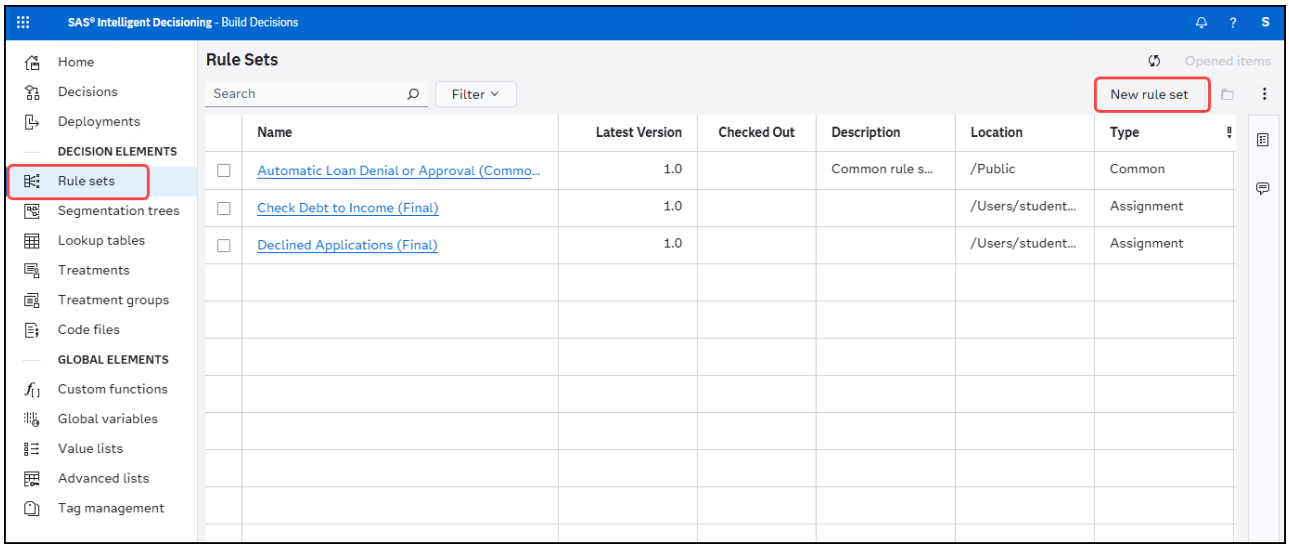


Create and Test an Assignment Rule Set

1. Select ➔ **Build Decisions** to open *SAS Intelligent Decisioning*.

2. Select to open the **Rule sets** page.

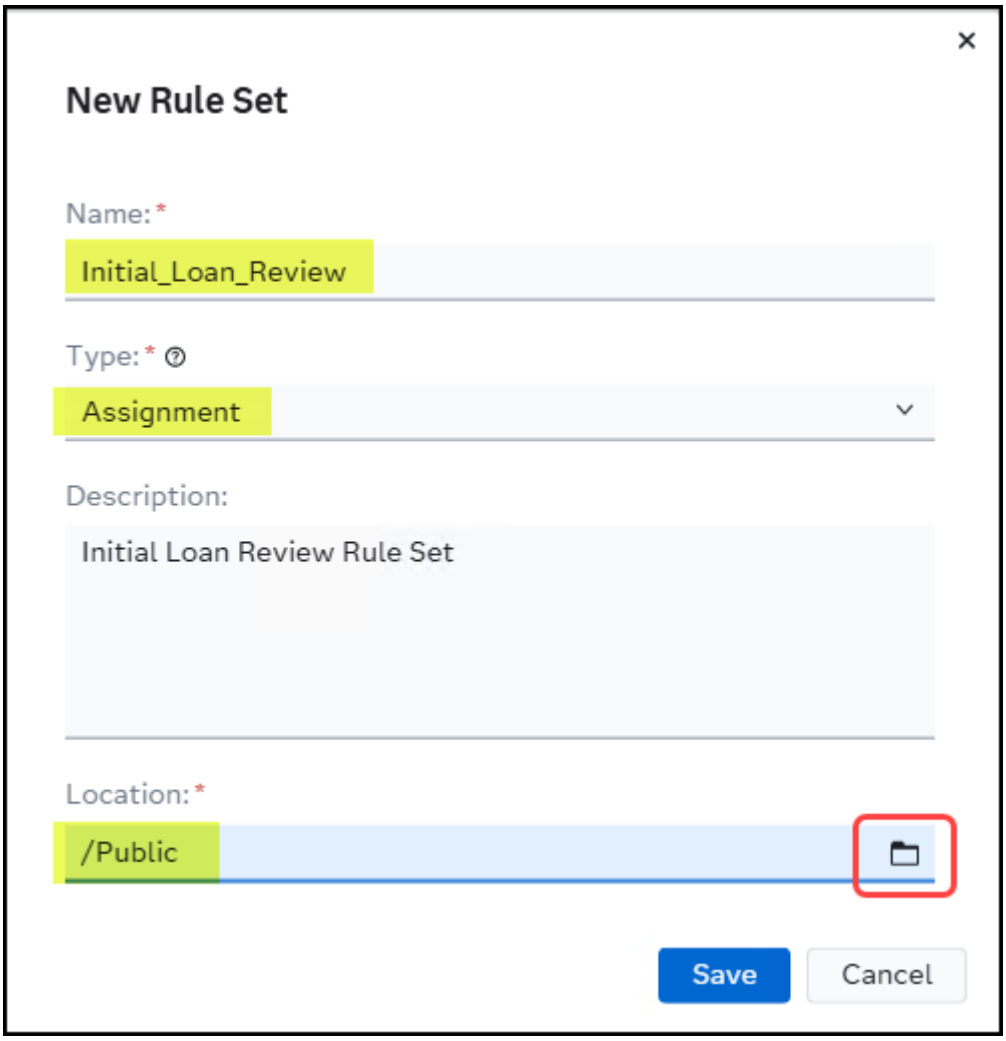
3. Click **New rule set** to create a new rule set.



4. Enter the following information for the new rule set:

   - Name: **Initial_Loan_Review**
   - Type: **Assignment** (default selection)
   - Description: **Initial Loan Review Rule Set**
   - Location: **/Public**

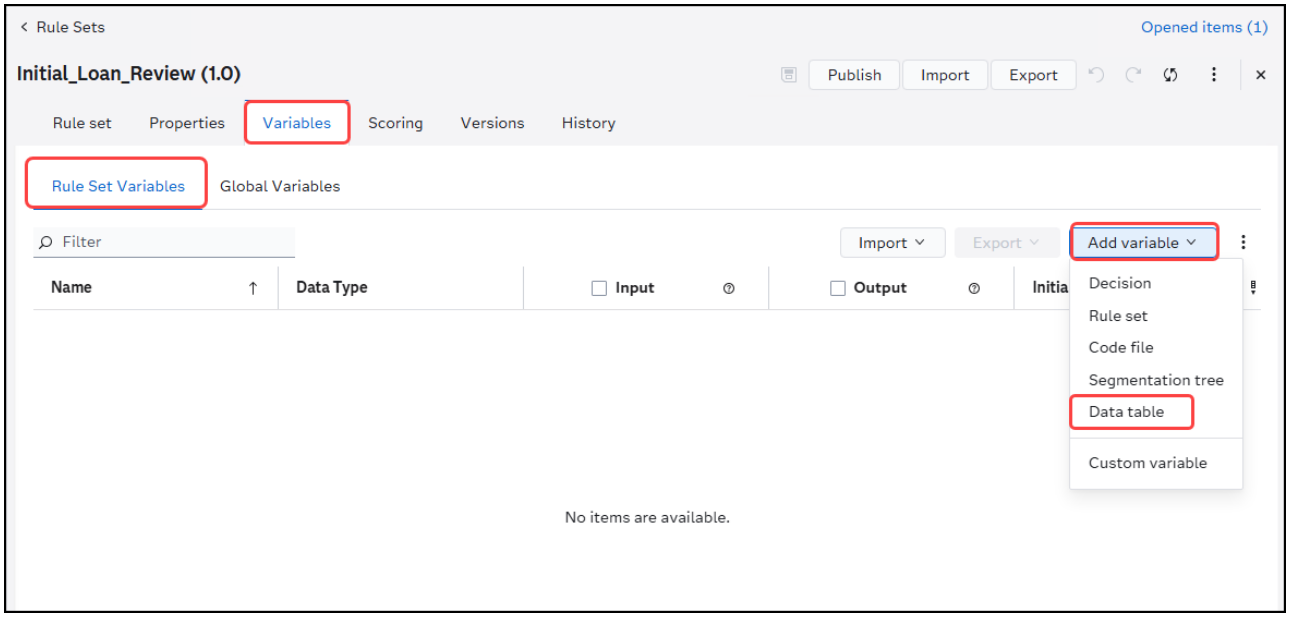
   > ✎ Select and navigate to **SAS Content ➔ Public**.



5. Click **Save** to create the rule set.

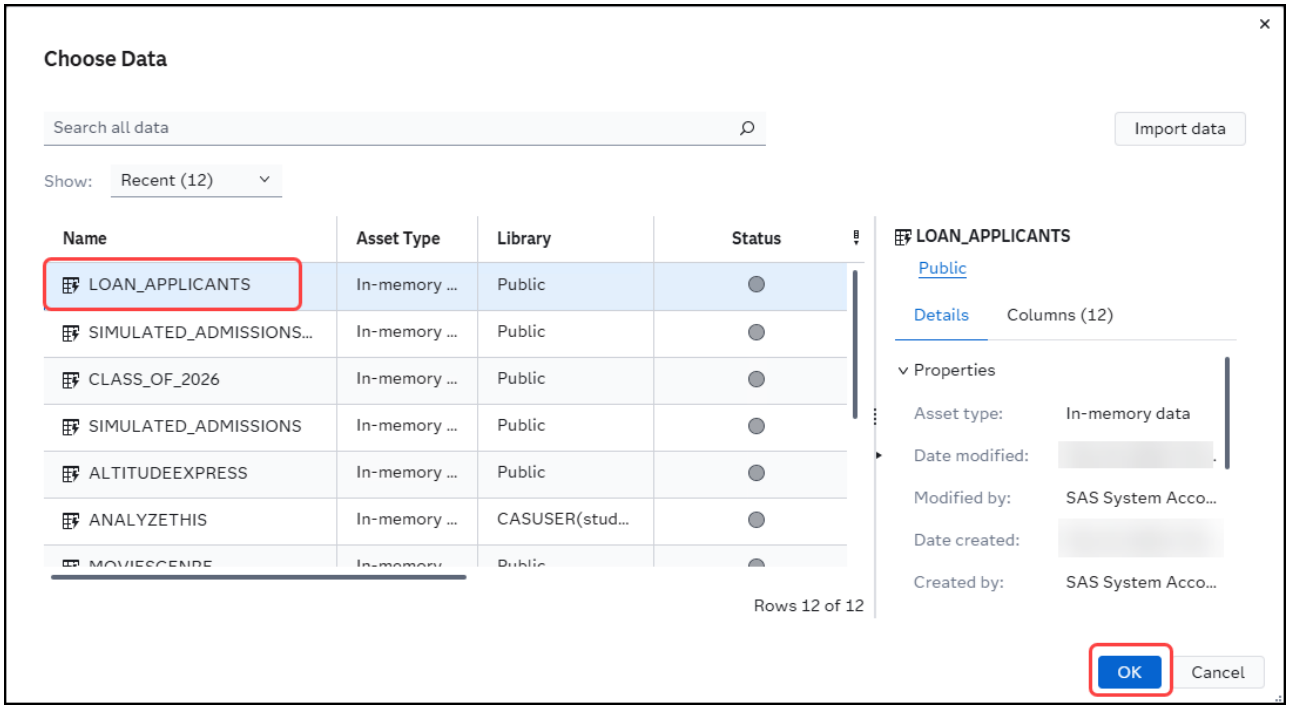
6. On the **Variables** tab, ensure that the **Rule Set Variables** sub-tab is selected.

7. Select **Add variable ➔ Data table** to add variables from an *in-memory* table.

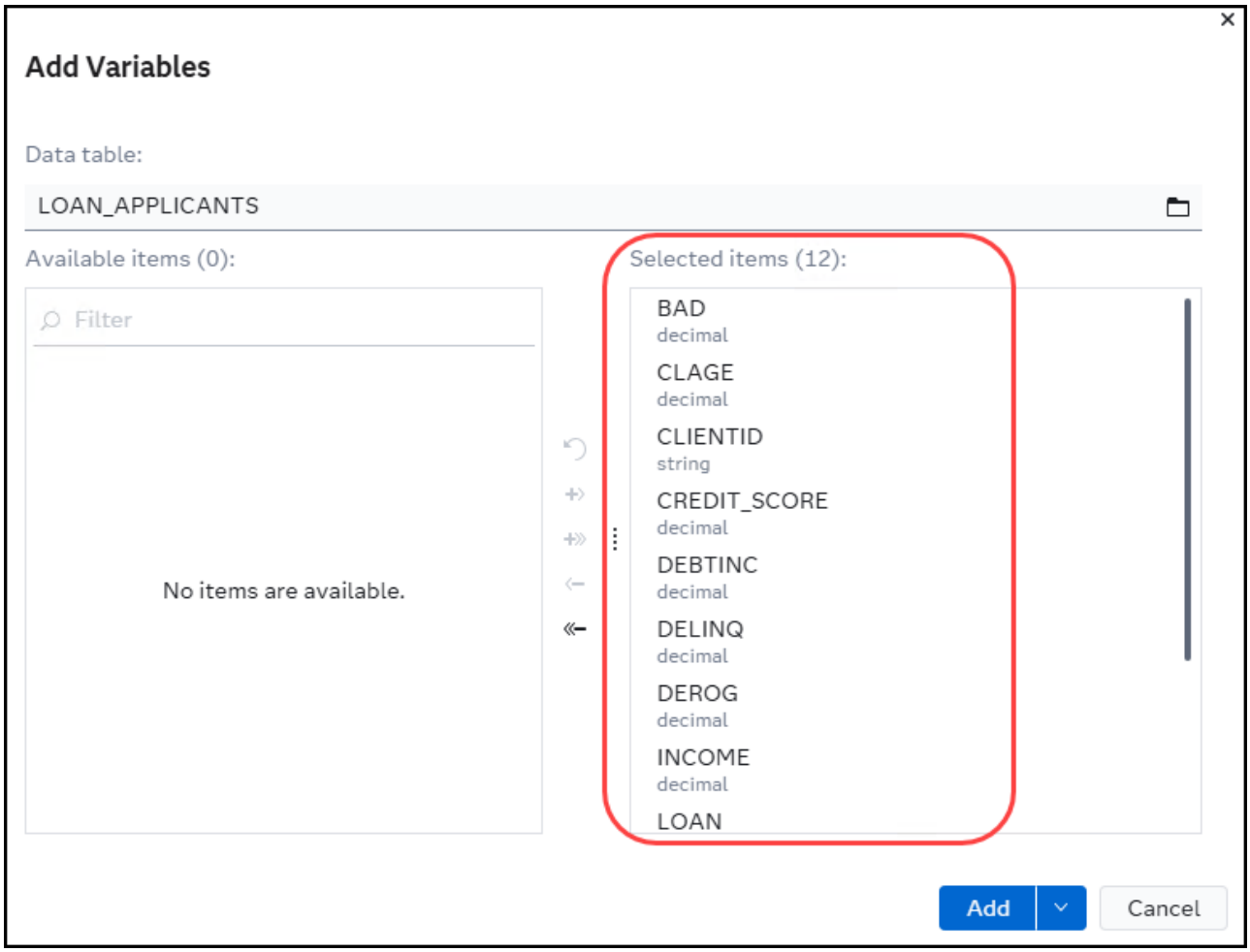


8. Select in the **Data table** selection.

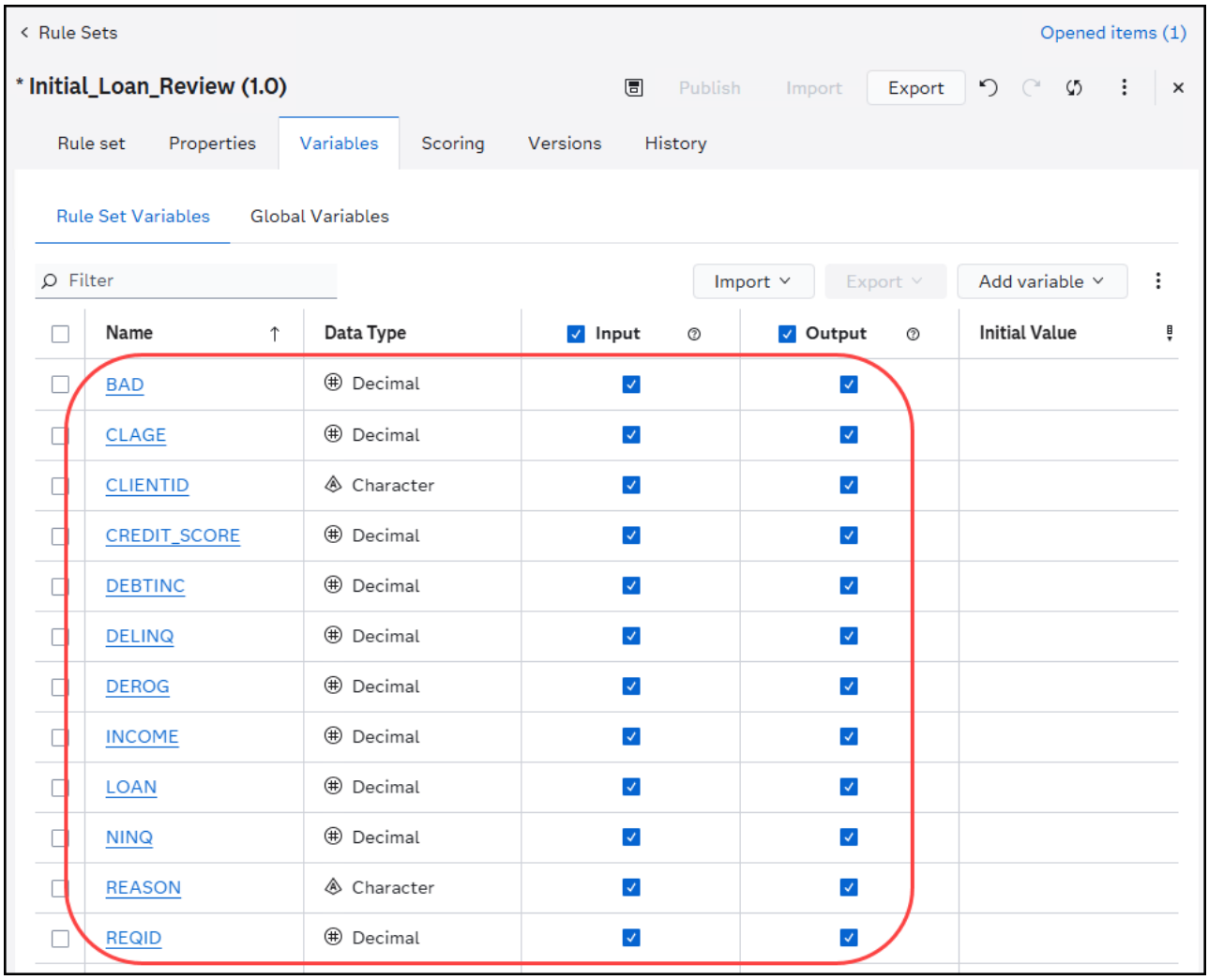
9. In the *Choose Data* dialog, select *Recent* from the drop-down menu next to *Show*, then select **LOAN_APPLICANTS** from the available data list and click **OK**.



10. Click to select all the columns from the table and add them to the selected items side.

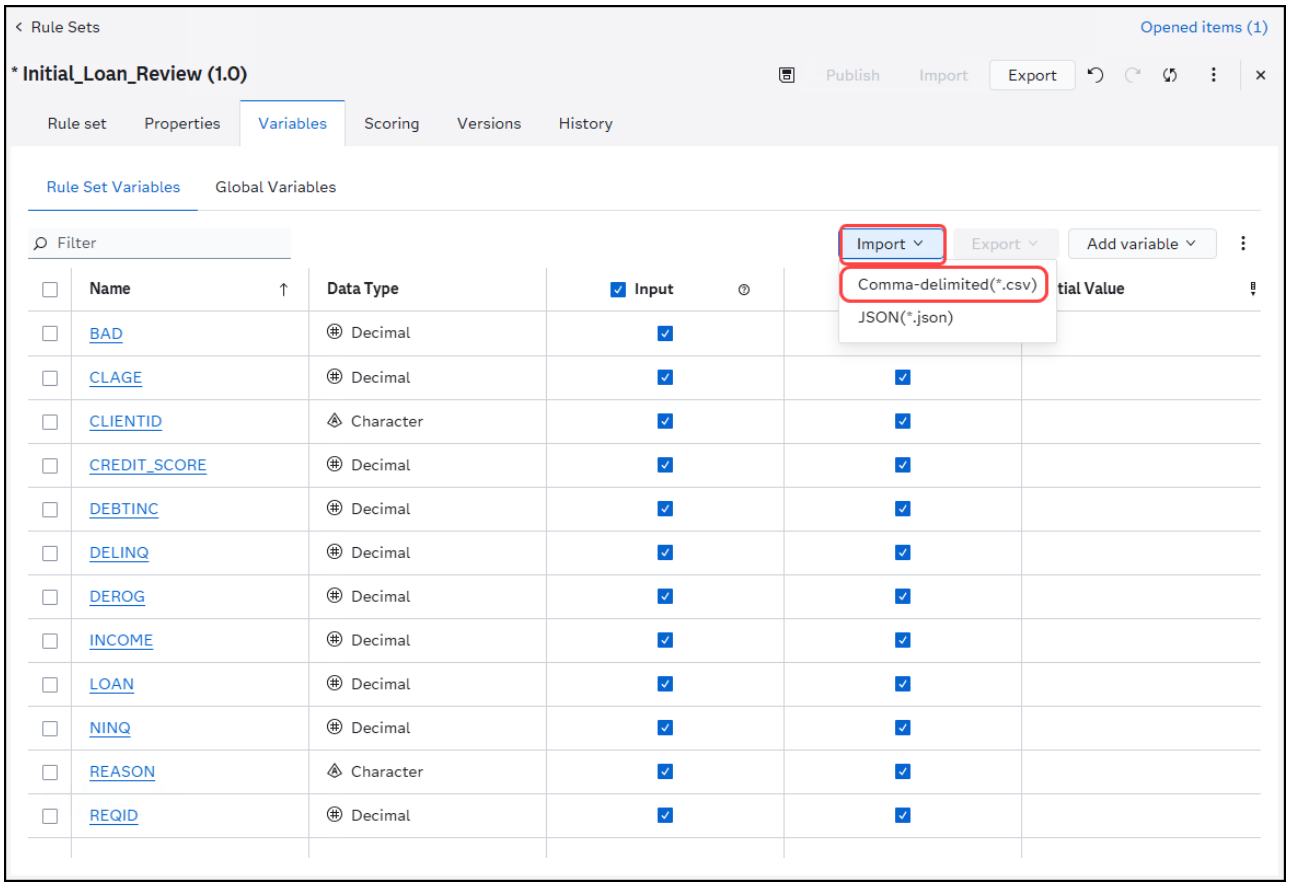


11. Click **Add** to add the selected items as variables for the rule set.



> ✎ The variables are automatically added as both *Input* and *Output* variables. You can uncheck those selections as appropriate for your rule set.

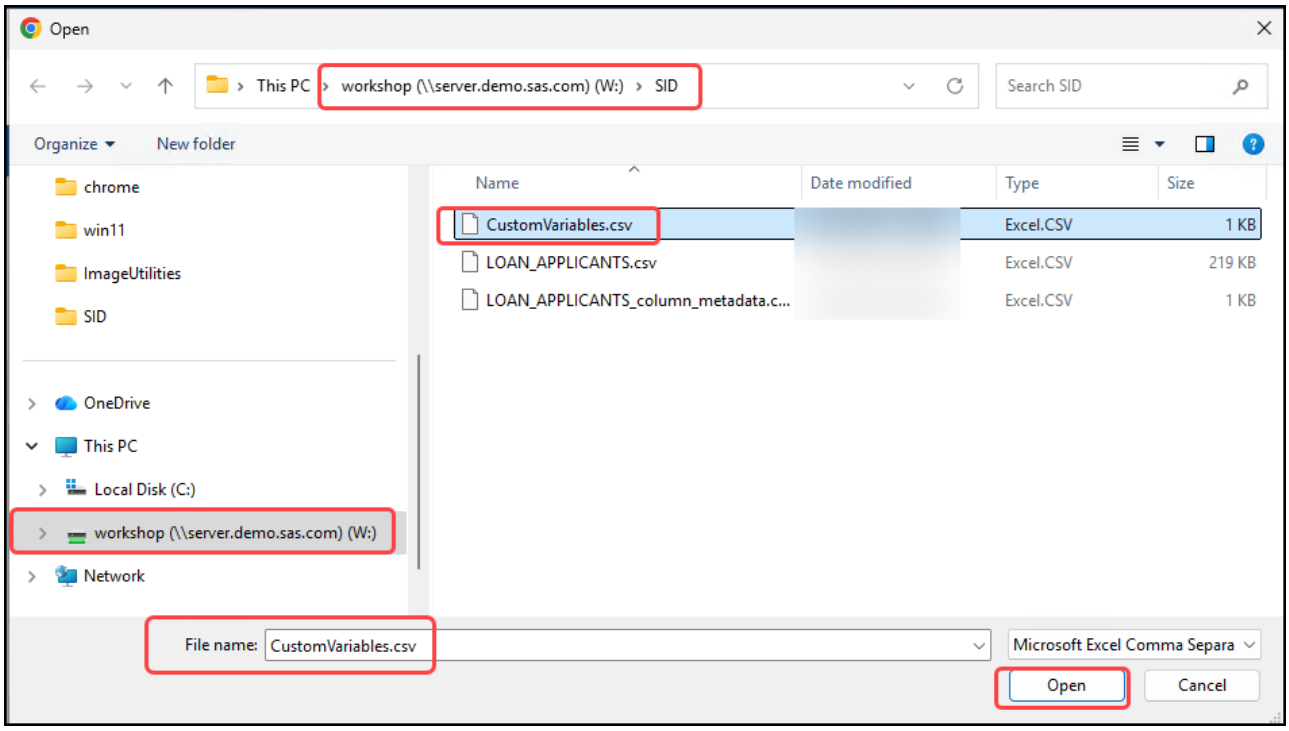
12. To import some custom output variables, select **Import ➔ Comma-delimited(*.csv)**.



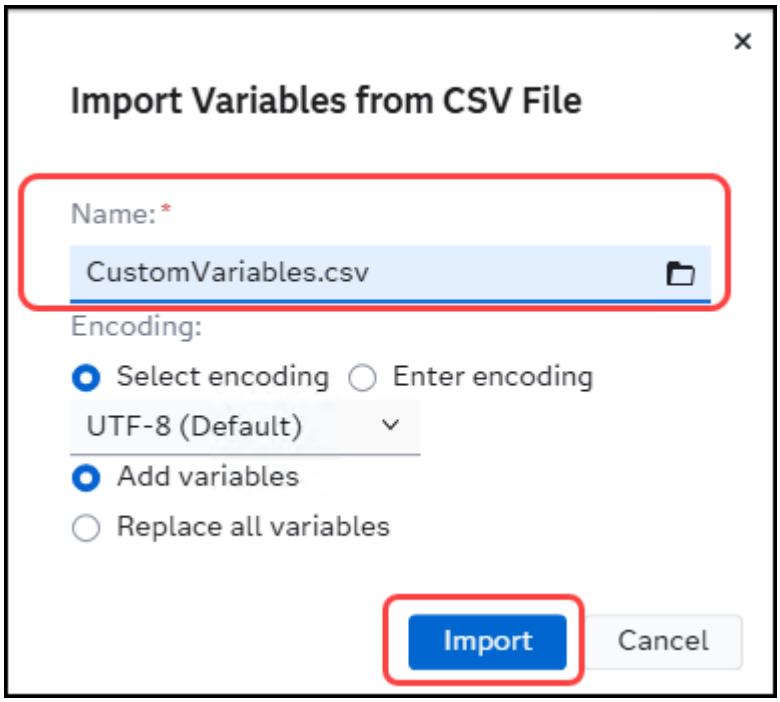
13. Select in the **Name** selection.

14. Navigate to **workshop (\\server.demo.sas.com) (W:)\SID**.

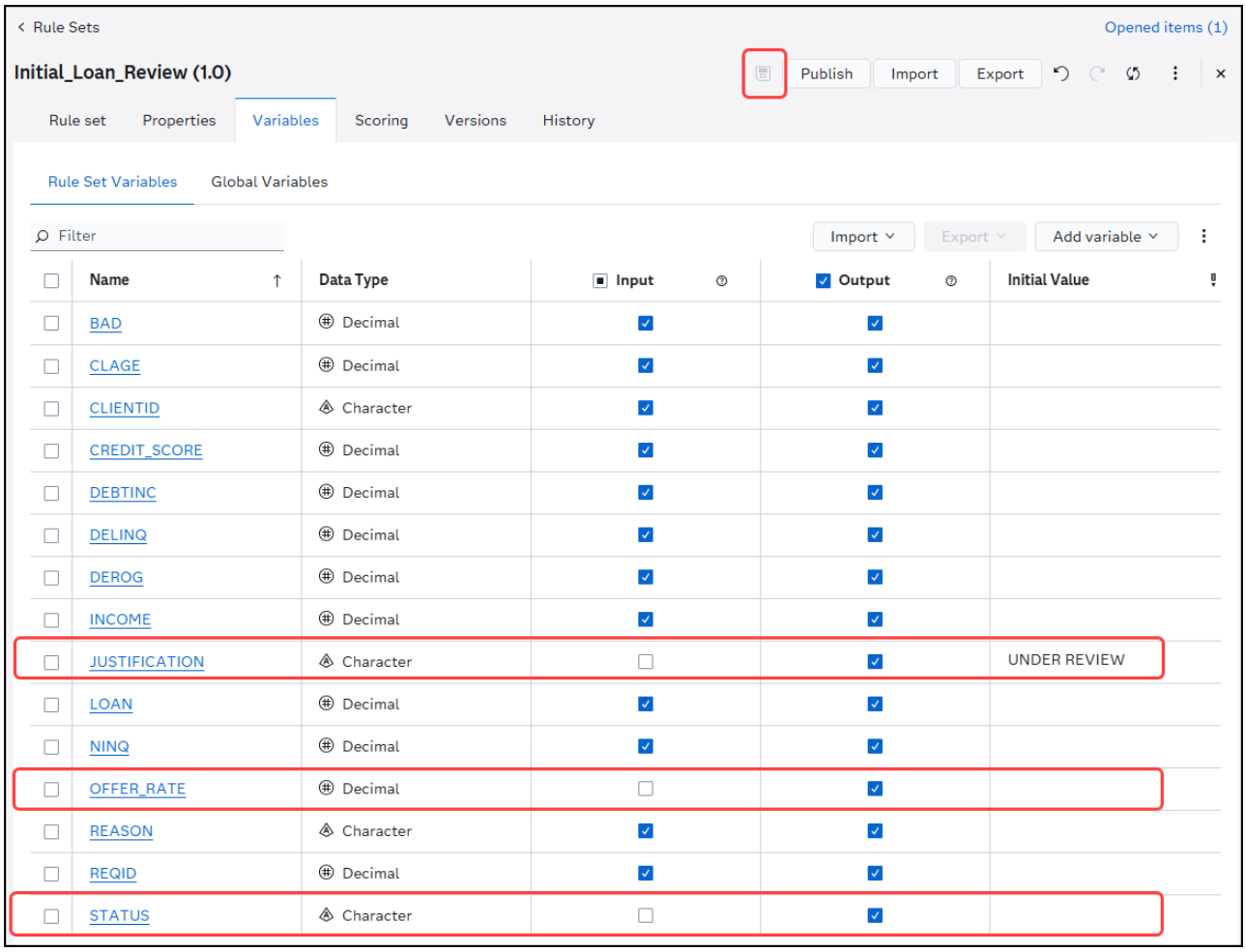
15. Select **CustomVariables.csv** and click **Open**.

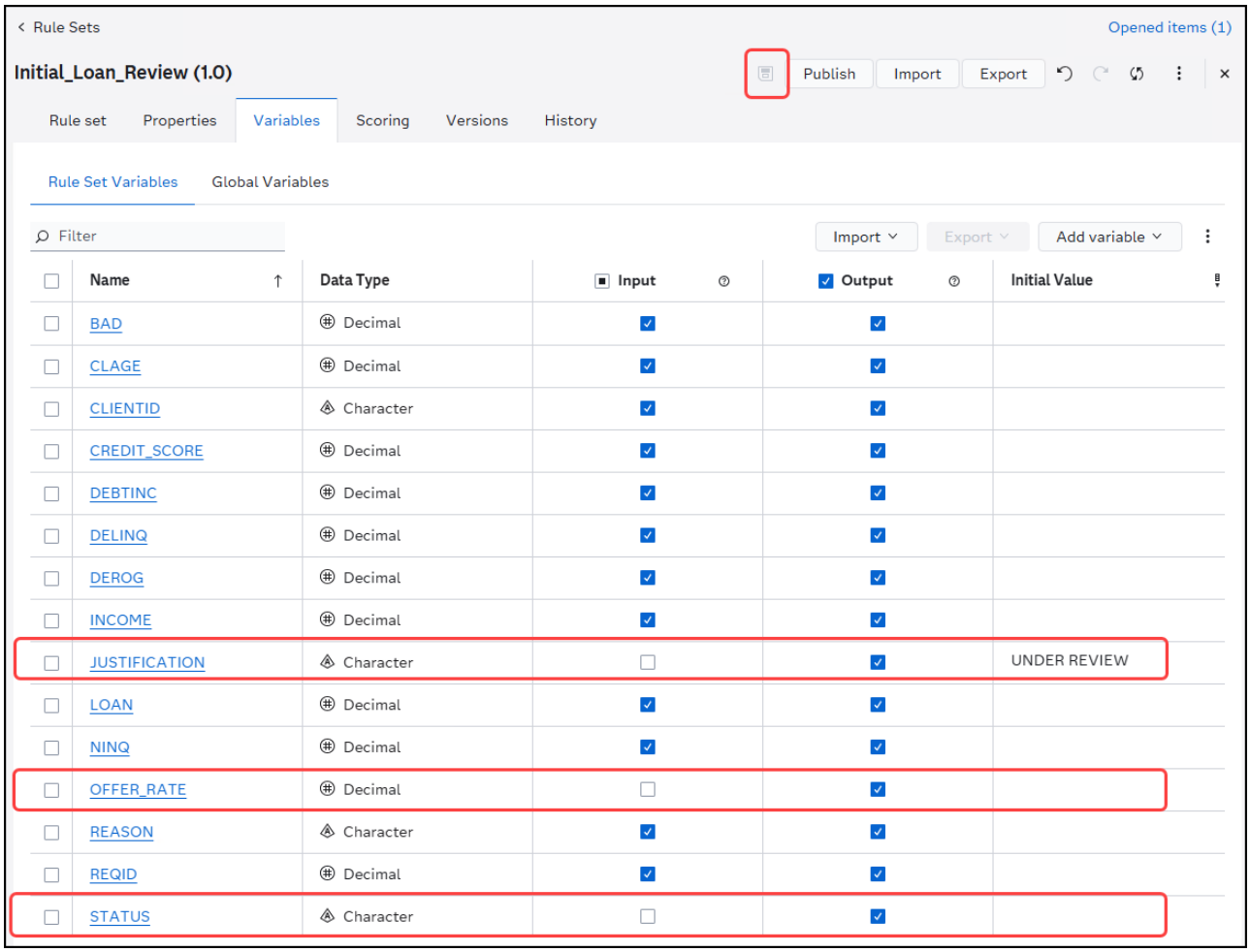


16. Click **Import** to import the variables from the selected CSV file.



17. Click to save the rule set.

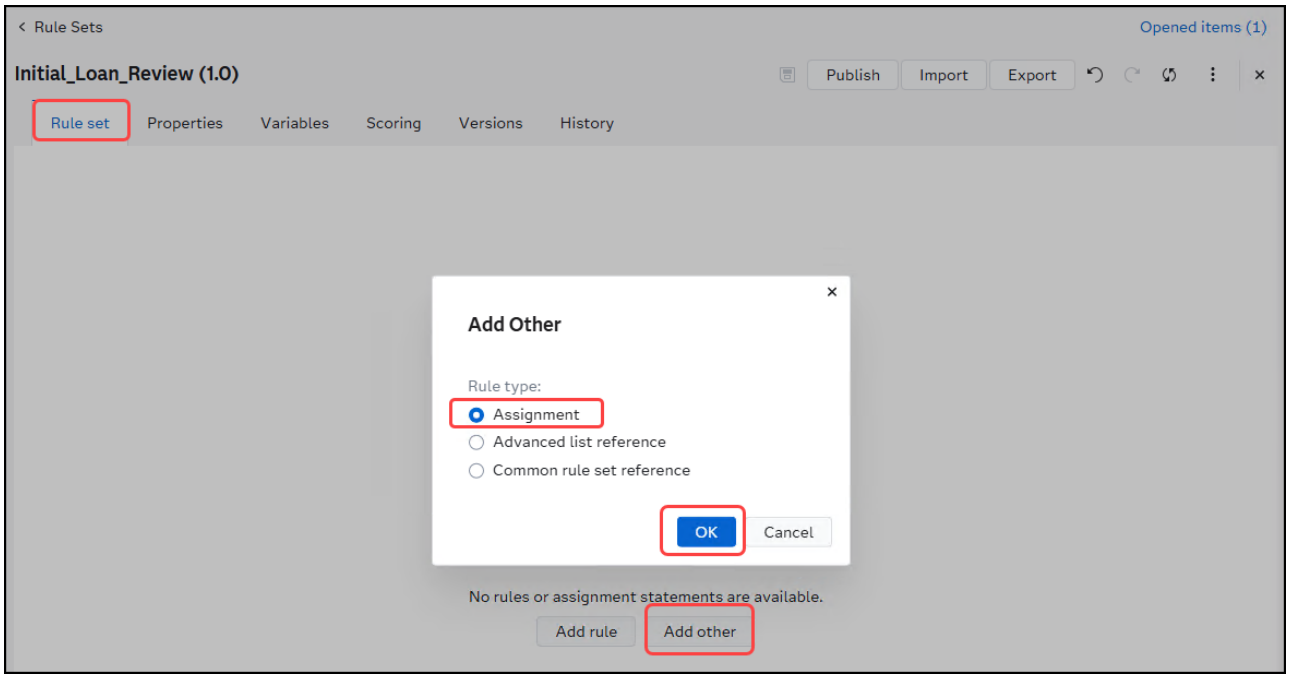




18. Select the **Rule set** tab.

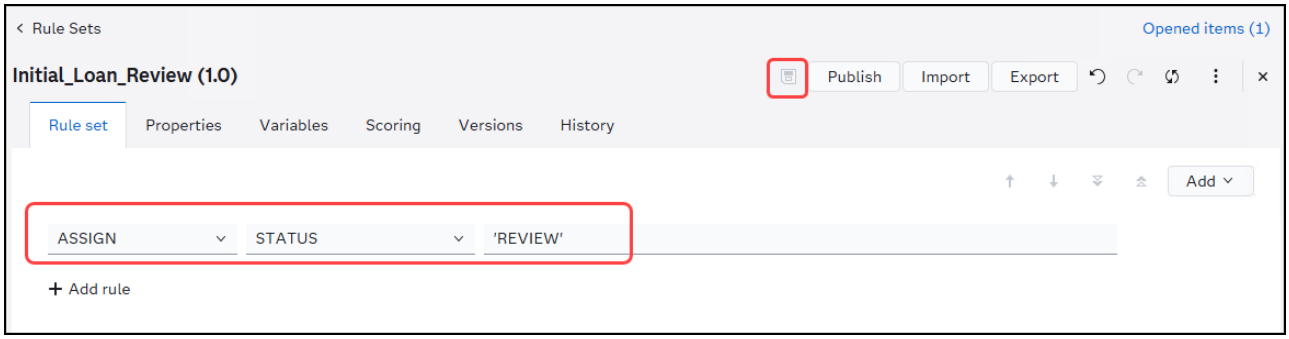
19. Click **Add other**.

20. Select **Assignment** for the *rule type* and click **OK**.

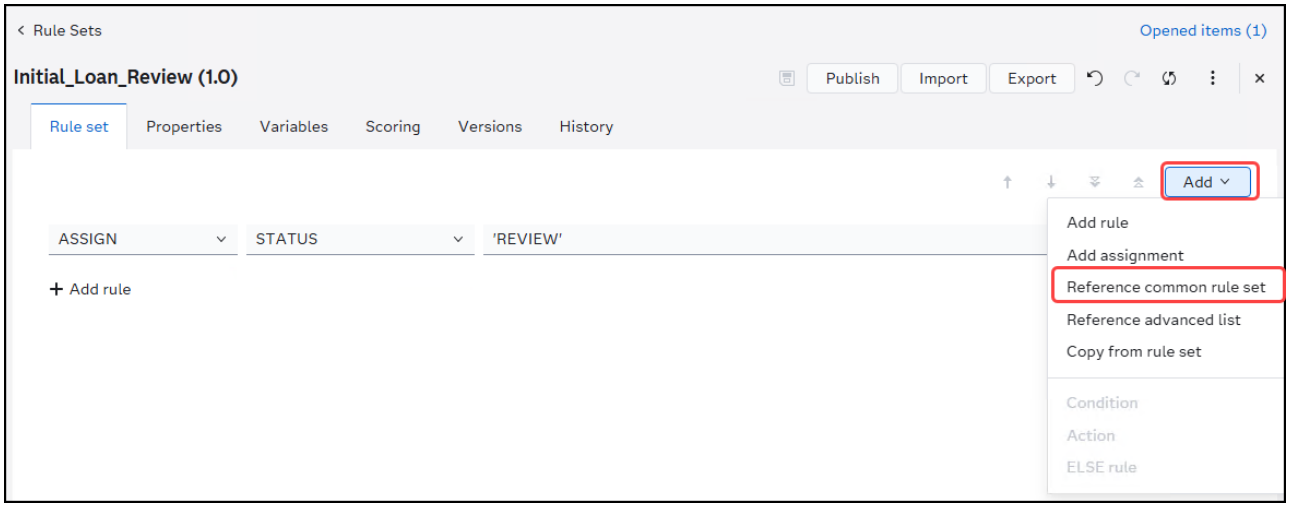


21. For the action assignment, select **ASSIGN STATUS 'REVIEW'**.

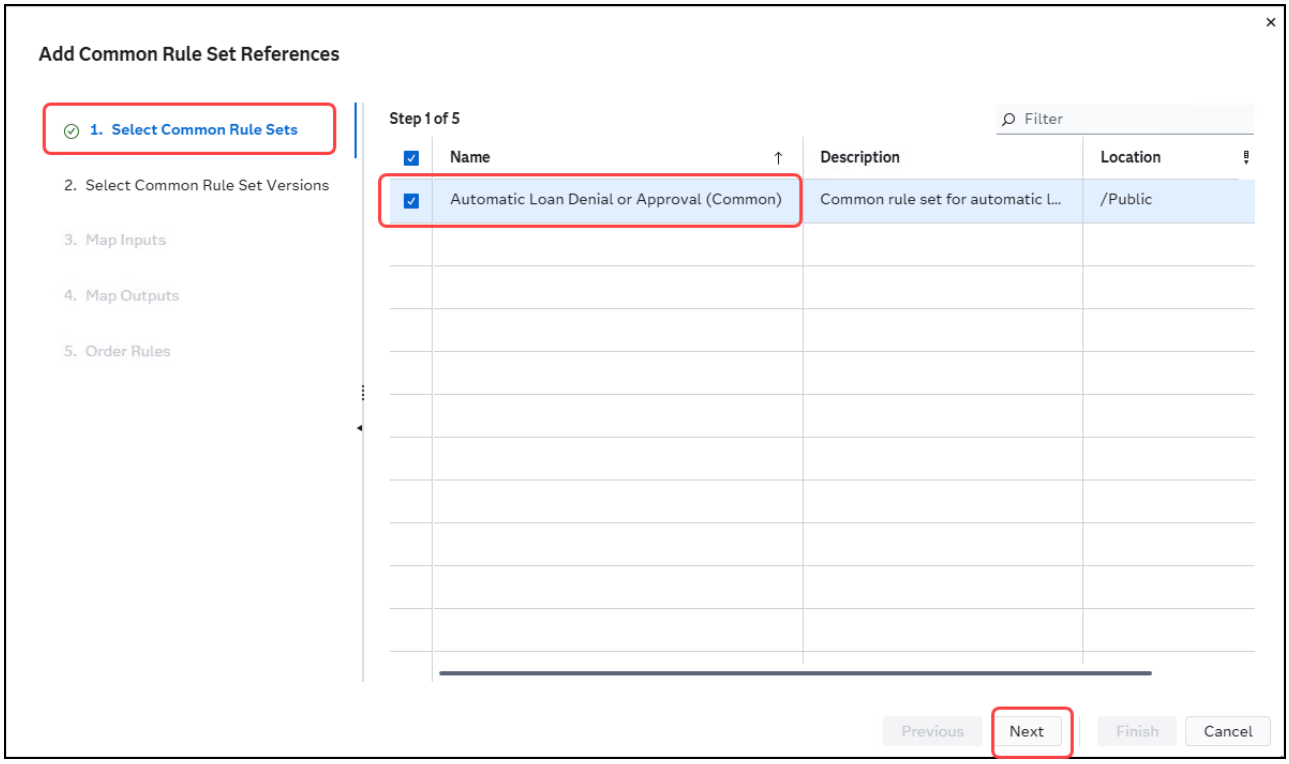
22. Click to save the rule set.



23. Select **Add ➔ Reference common rule set** to add a common rule set that has already been created.



24. On the *Select Common Rule Sets* tab, check the common rule set **Automatic Loan Denial or Approval (Common)** and click **Next**.



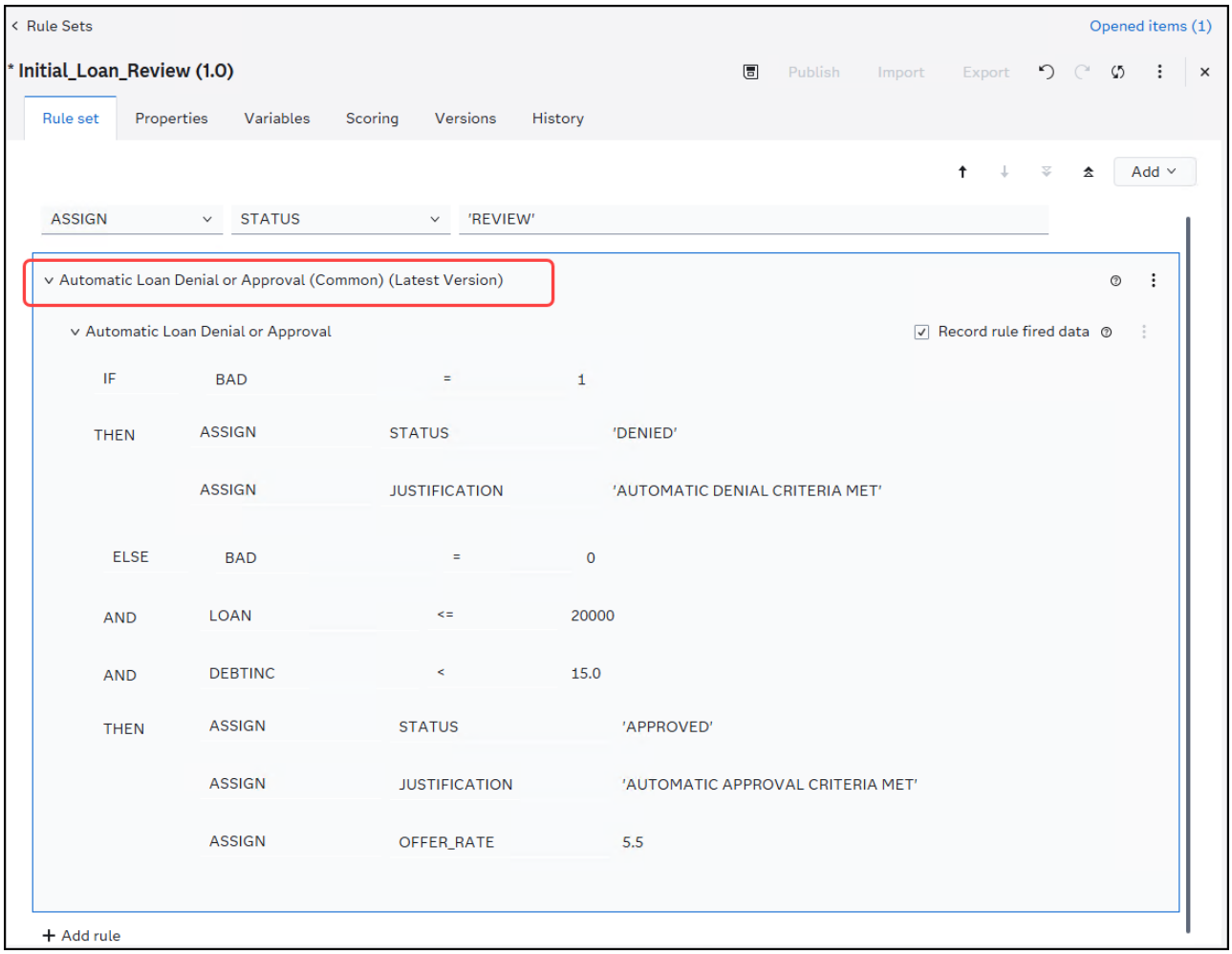
25. On the *Select Common Rule Set Versions* tab, click **Next**.

26. On the *Map Inputs* tab, the inputs are auto-mapped since the same input variable names exist in both rule sets. Click **Next**.

27. On the *Map Outputs* tab, the inputs are auto-mapped since the same output variable names exist in both rule sets. Click **Next**.

28. On the *Order Rules* tab, click **Finish** to add the selected common rule set(s) to the current rule set.

29. Expand the **Automatic Loan Denial or Approval (Common) (Latest Version)** rule to view its logic.



30. Click to save the rule set.

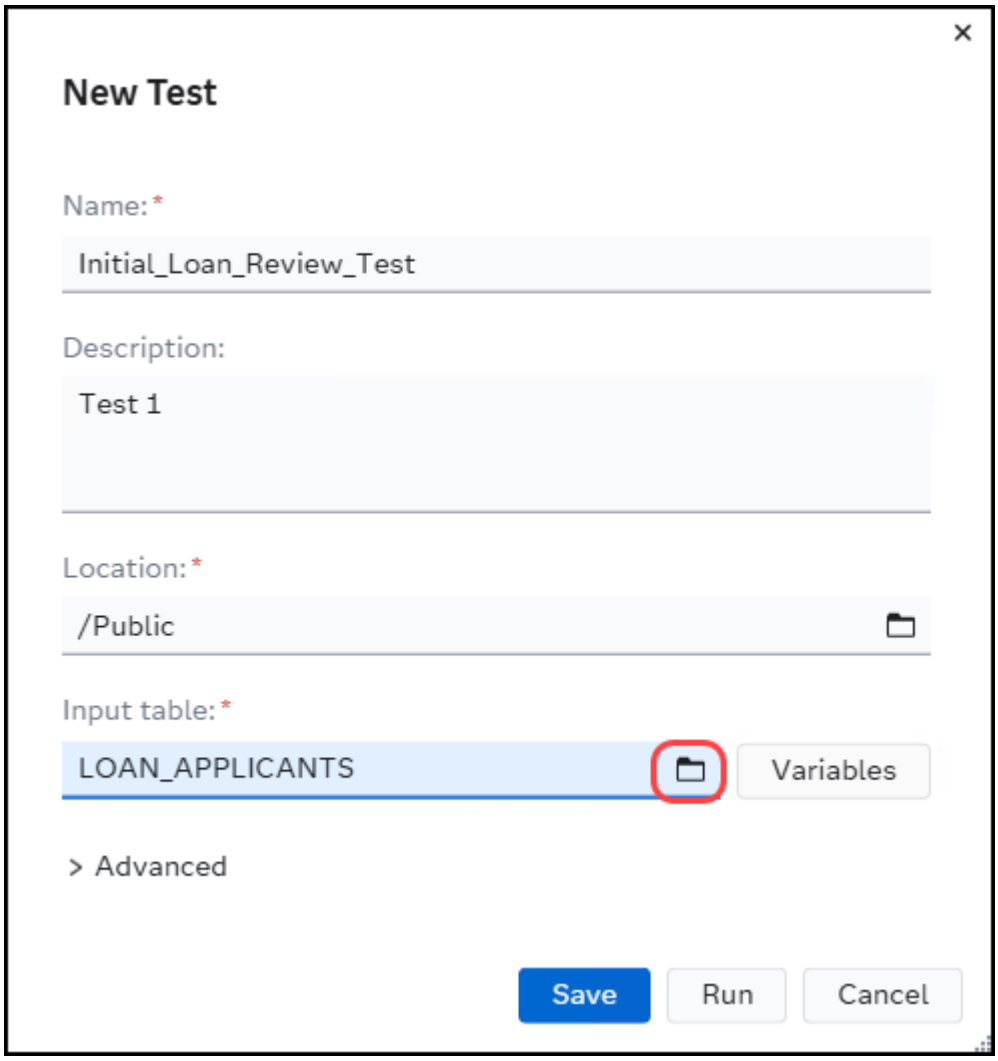
31. To test the rule set, select the **Scoring** tab.

32. On the **Tests** sub-tab, click **New Test**.

33. Enter the following information:

    - Name: **Initial_Loan_Review_Test**
    - Description: **Test 1**
    - Location: **/Public**
    - Input table: **LOAN_APPLICANTS**.

    > ✎ Click to select the *Input table*.



34. Click **Variables** to view the variables mapping between the input table and the rule set variables.

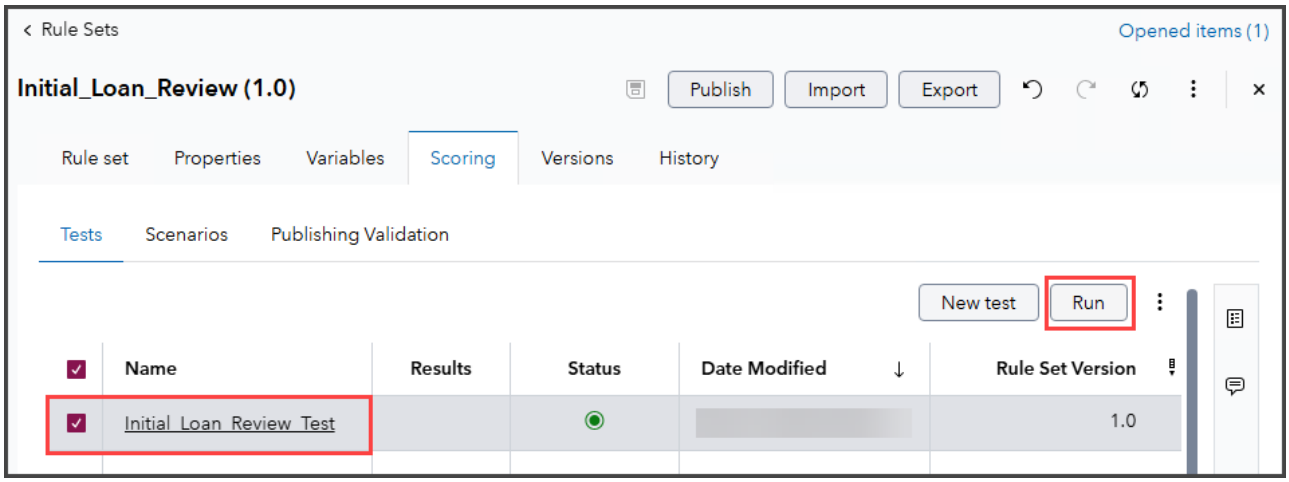
> ✎ All the variables are automatically mapped since the variable names on the rule set match the names on the input table.

35. Click **OK** to close the *Variable Mappings* dialog.

36. Click **Save** to save this test to the rule set.

> ✎ If you do not select **Save** and simply select **Run**, then the test is run; however, the test is not saved to the rule set and if your session is closed, you will need to set up the test again to run it in the future.

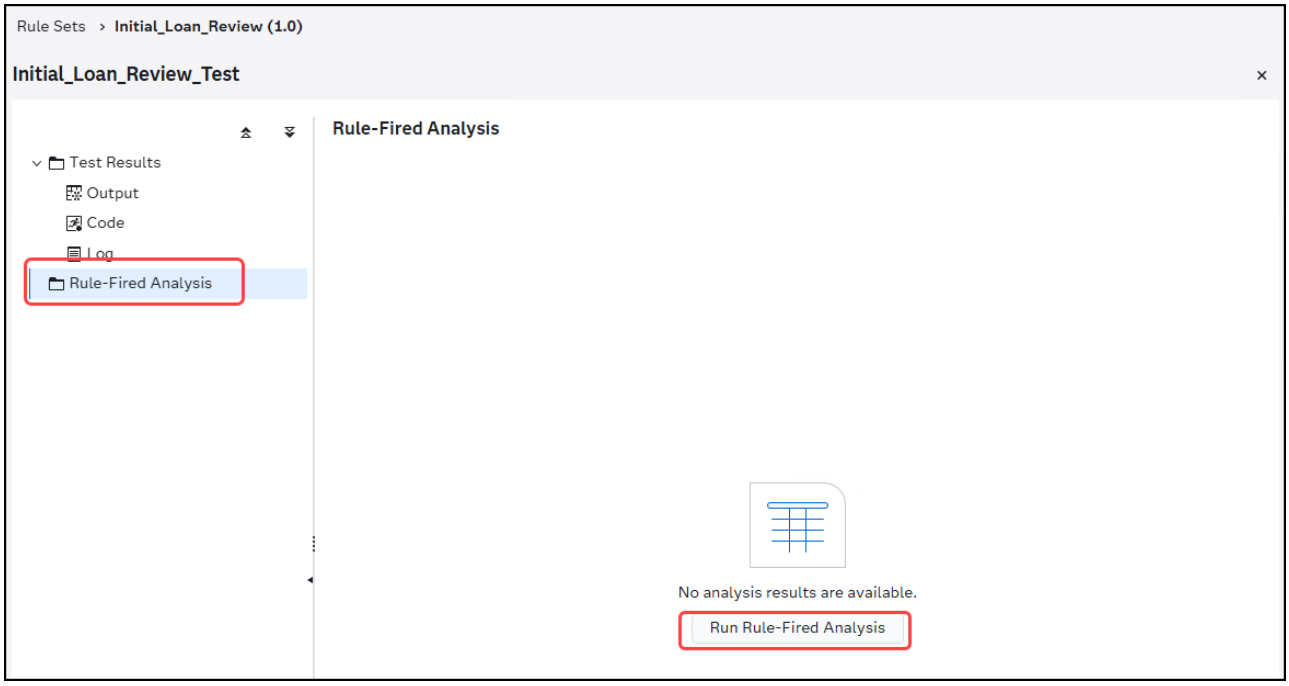
37. Check the newly created test and click **Run** to run the specified test.



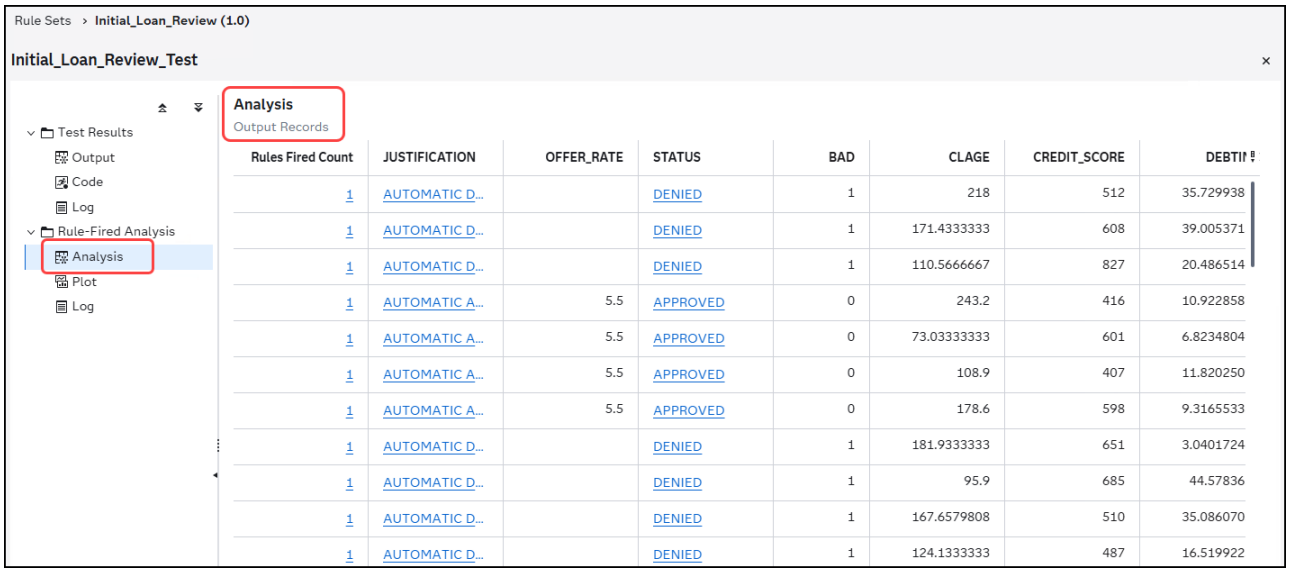
> ✎ It may take a few seconds for the test to complete. You can select to refresh the test status.

38. Once the test run is complete, select to view the test results.

39. Select **Rule-Fired Analysis** and then click **Run Rule-Fired Analysis** to view which records triggered a rule.



40. Once the rule-fired analysis has run, review the results on the *Analysis* tab.



Rules Fired Count	JUSTIFICATION	OFFER_RATE	STATUS	BAD	CLAGE	CREDIT_SCORE	DEBTI
1	AUTOMATIC D...		DENIED	1	218	512	35.729938
1	AUTOMATIC D...		DENIED	1	171.4333333	608	39.005371
1	AUTOMATIC D...		DENIED	1	110.5666667	827	20.486514
1	AUTOMATIC A...	5.5	APPROVED	0	243.2	416	10.922858
1	AUTOMATIC A...	5.5	APPROVED	0	73.03333333	601	6.8234804
1	AUTOMATIC A...	5.5	APPROVED	0	108.9	407	11.820250
1	AUTOMATIC A...	5.5	APPROVED	0	178.6	598	9.3165533
1	AUTOMATIC D...		DENIED	1	181.9333333	651	3.0401724
1	AUTOMATIC D...		DENIED	1	95.9	685	44.57836
1	AUTOMATIC D...		DENIED	1	167.6579808	510	35.086070
1	AUTOMATIC D...		DENIED	1	124.1333333	487	16.519922

41. Click **x** close the test.

42. Click **x** to close the rule set.

Create and Test a DS2 Code File

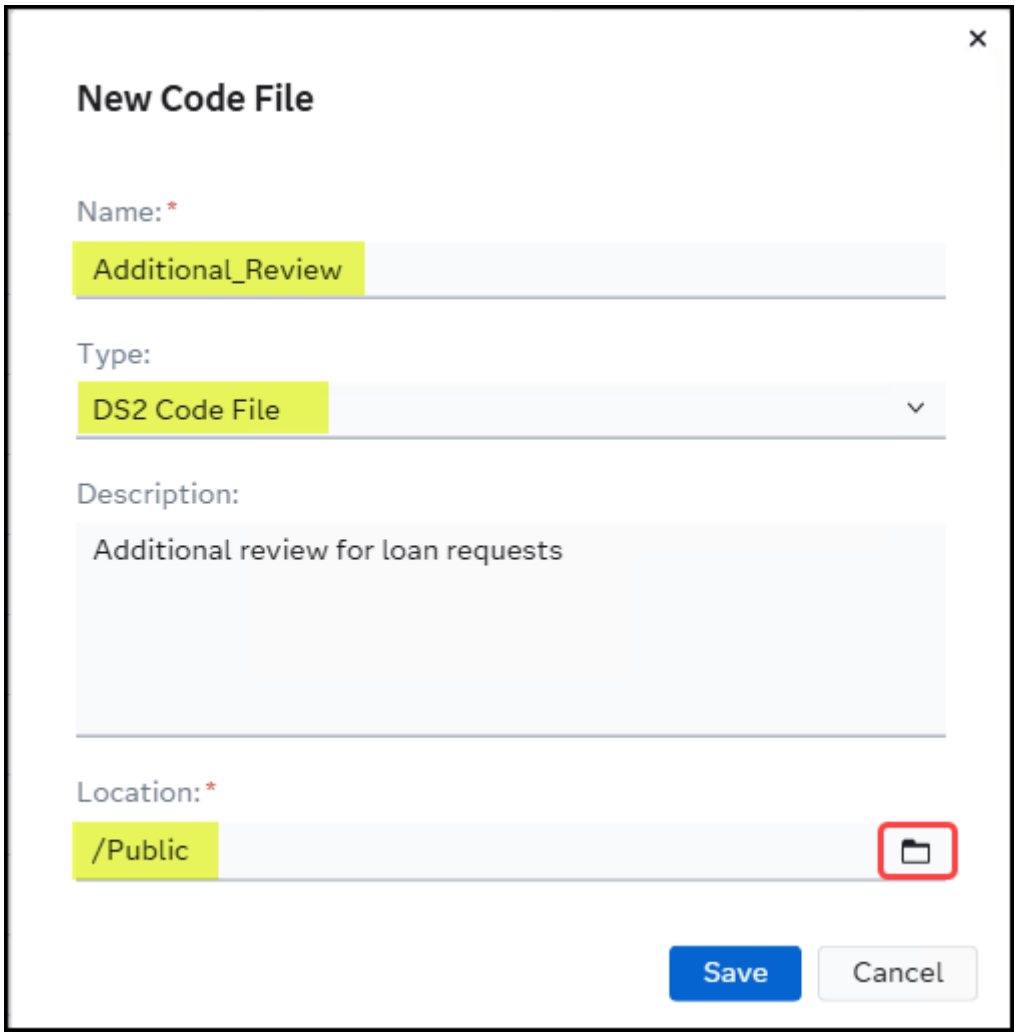
1. Select to open the **Code files** page.

2. Click **New code file** to create a new code file.

3. Enter the following information:

    - Name: **Additional_Review**
    - Type: **DS2 Code File** (default selection)
    - Description: **Additional review for loan requests**
    - Location: **/Public**.

> ☛ Select to navigate to this folder in *SAS Content*.



4. Click **Save** to create the DS2 Code file.

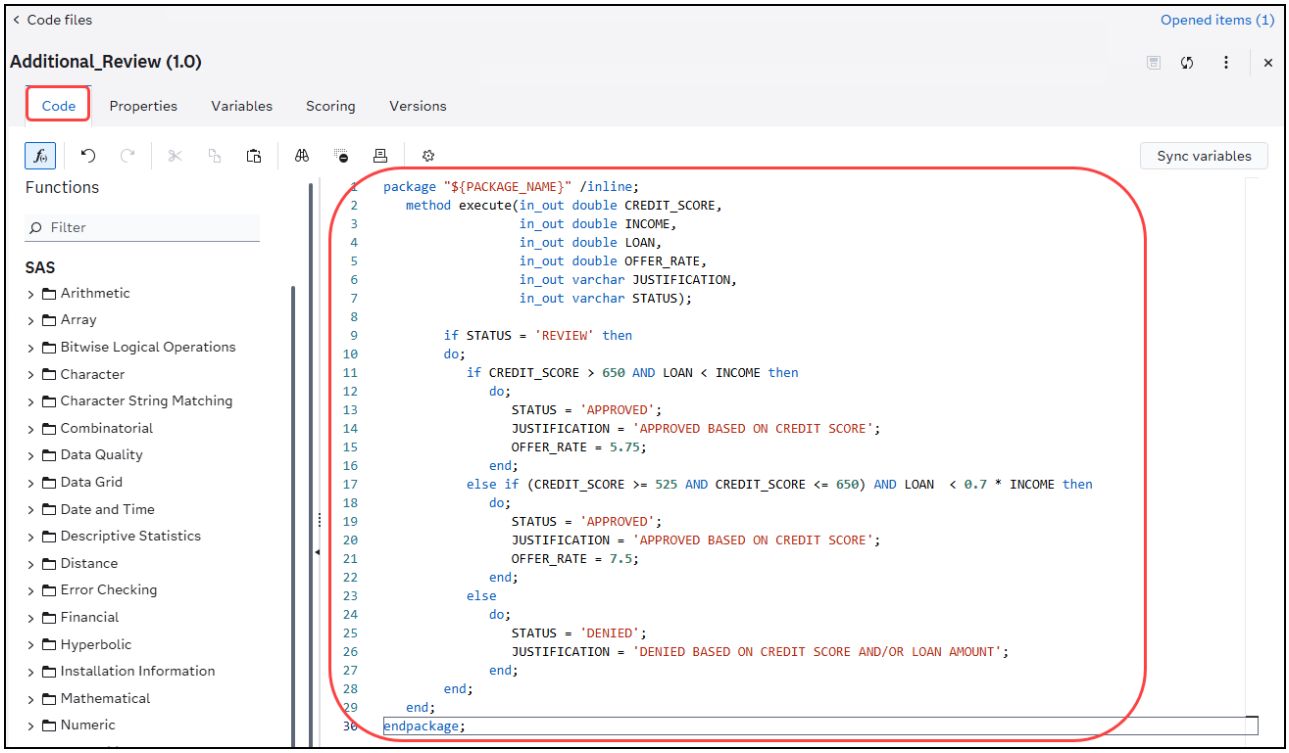
5. Replace the default code with the code block below:

```
package "${PACKAGE_NAME}" /inline;
   method execute(in_out double CREDIT_SCORE,
                  in_out double INCOME,
                  in_out double LOAN,
                  in_out double OFFER_RATE,
                  in_out varchar JUSTIFICATION,
                  in_out varchar STATUS);

        if STATUS = 'REVIEW' then
        do;
           if CREDIT_SCORE > 650 AND LOAN < INCOME then
              do;
                 STATUS = 'APPROVED';
                 JUSTIFICATION = 'APPROVED BASED ON CREDIT SCORE';
                 OFFER_RATE = 5.75;
              end;
           else if (CREDIT_SCORE >= 525 AND CREDIT_SCORE <= 650) AND LOAN  <
0.7 * INCOME then
              do;
                 STATUS = 'APPROVED';
                 JUSTIFICATION = 'APPROVED BASED ON CREDIT SCORE';
```

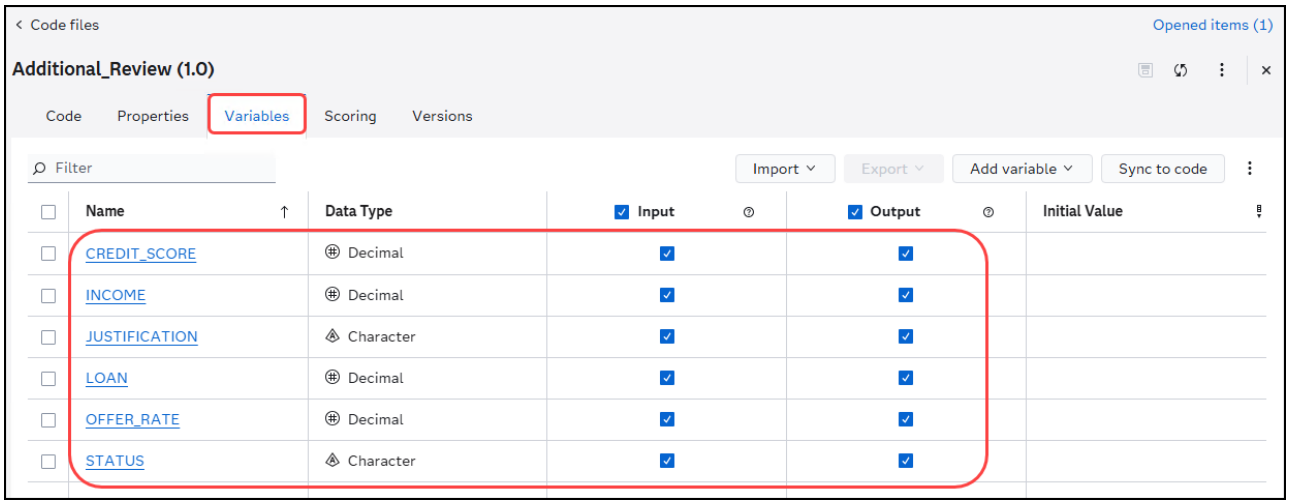
```
                    OFFER_RATE = 7.5;
                end;
            else
                do;
                    STATUS = 'DENIED';
                    JUSTIFICATION = 'DENIED BASED ON CREDIT SCORE AND/OR LOAN
AMOUNT';
                end;
        end;
    end;
endpackage;
```

6. Click to save the code file.



7. Click **Sync variables** to add the variables declared in the DS2 code to the *Variables* tab for for the code file.

8. Select the **Variables** tab to confirm the variables were added.

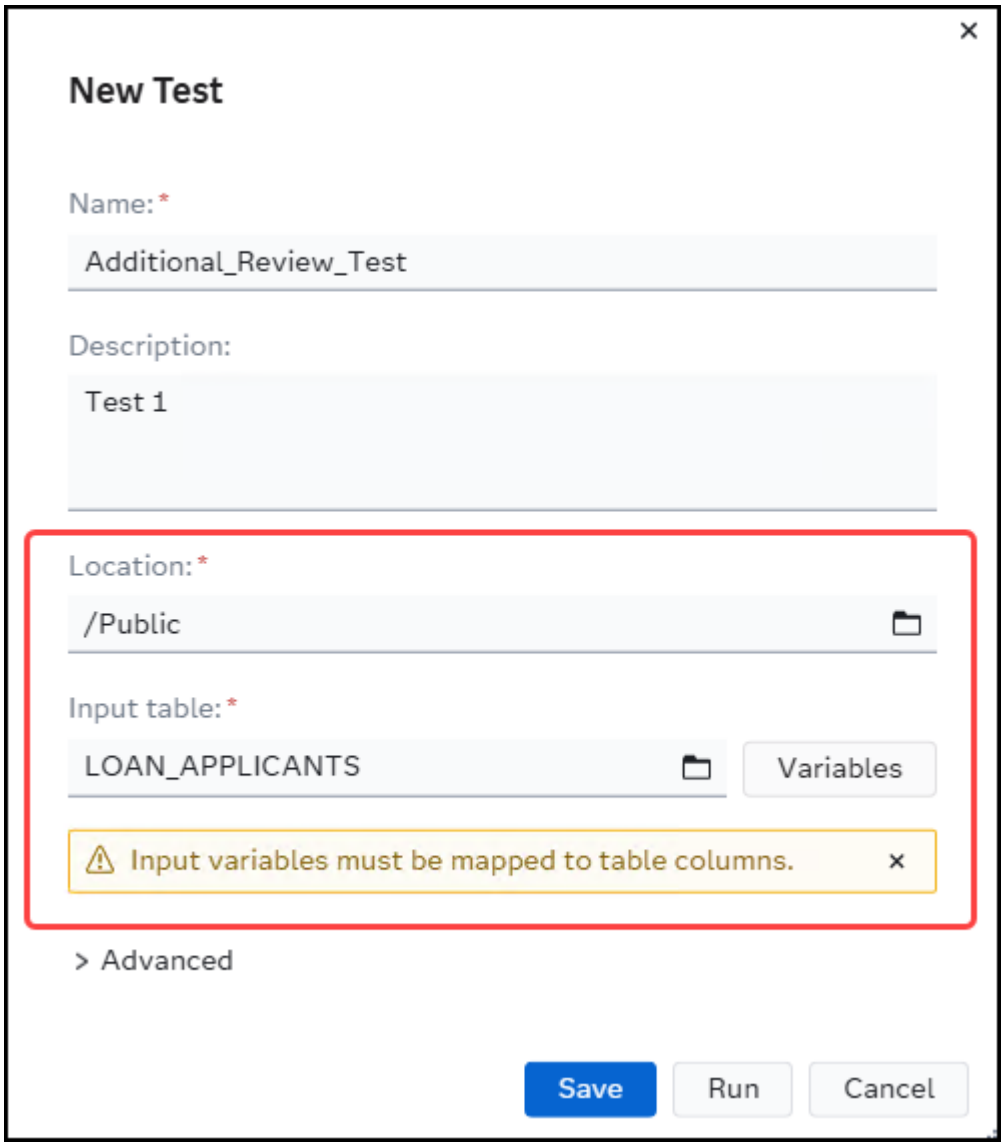


9. To test the code file, select the **Scoring** tab.

10. On the **Tests** sub-tab, click **New Test**.

11. Enter the following information:

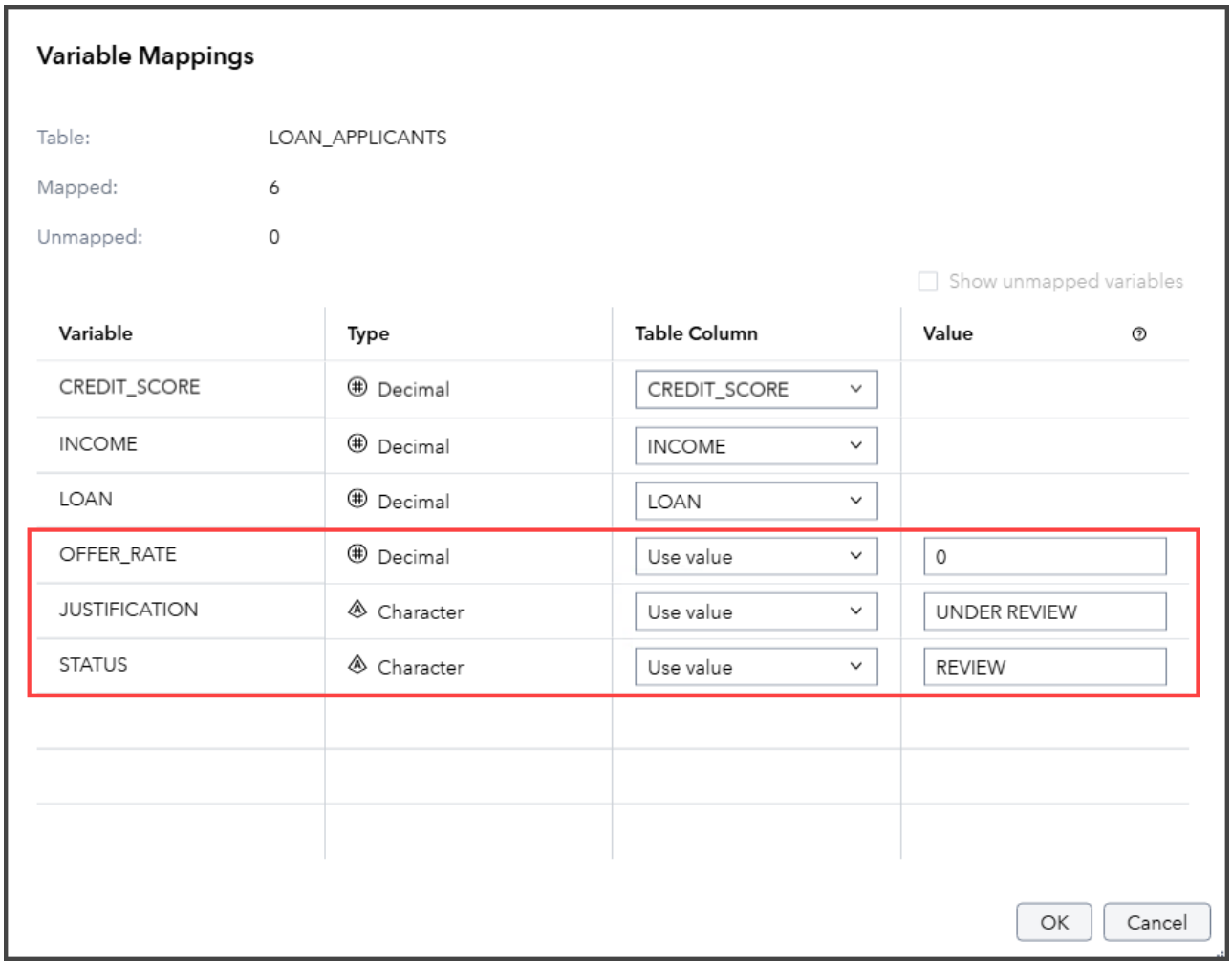
    - Name: **Additional_Review_Test**
    - Description: **Test 1**
    - Location: **/Public**
    - Input table: **LOAN_APPLICANTS**.



12. Click **Variables** to resolve the input variable mapping.

13. Enter the following information for unmapped variables:

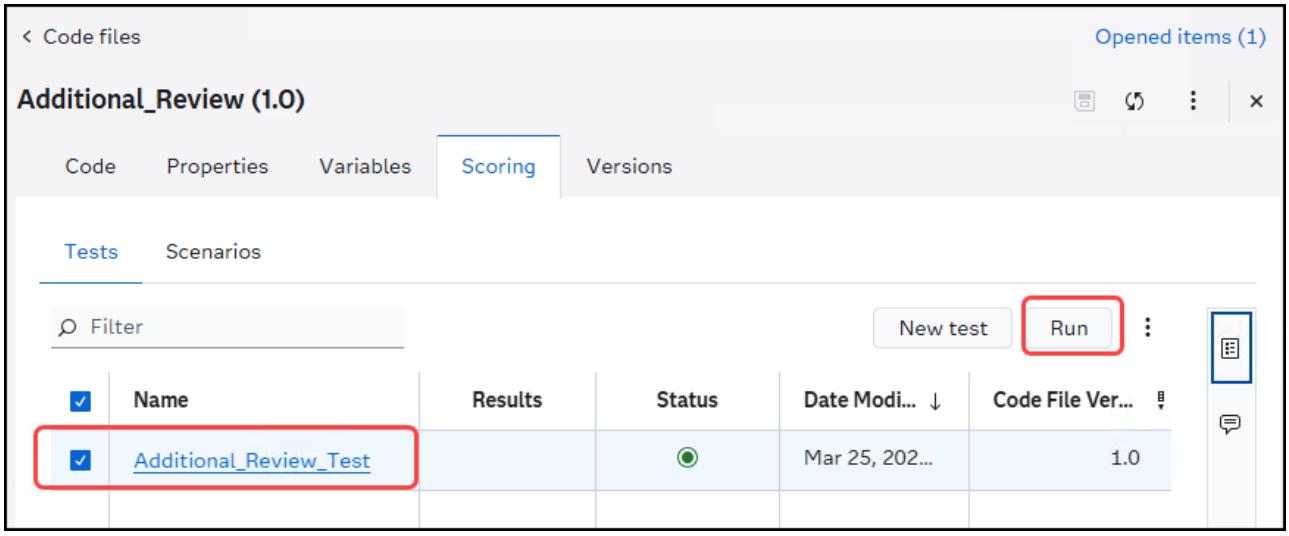
Variable	Type	Table Column	Value
OFFER_RATE	Decimal	**Use value**	**0**
JUSTIFICATION	Character	**Use value**	**UNDER REVIEW**
STATUS	Character	**Use value**	**REVIEW**



14. Click **OK** to close the *Variable Mappings* dialog.

15. Click **Save** to save this test to the code file.

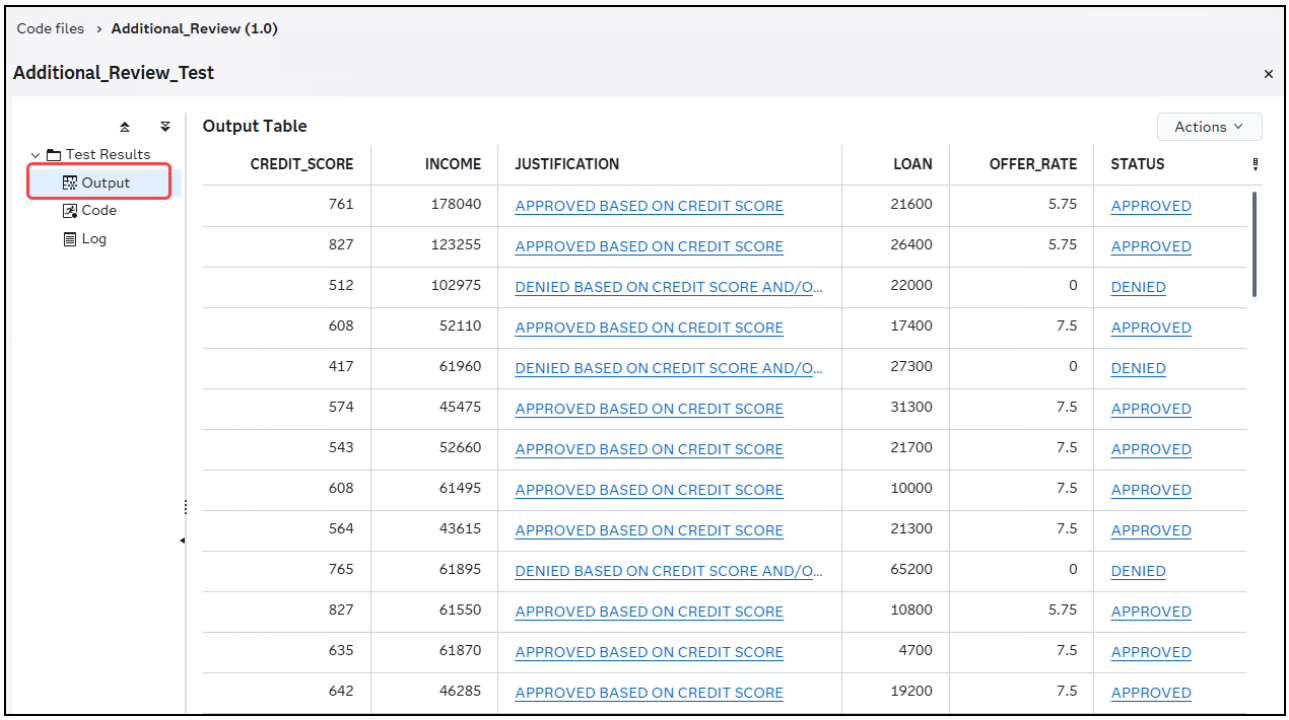
16. Check the newly created test and click **Run** to run the specified test.



> ✎ It may take a few seconds for the test to complete. You can select to refresh the test status.

17. Once the test run is complete, select to view the test results.

18. Review the results on the **Output** tab.



CREDIT_SCORE	INCOME	JUSTIFICATION	LOAN	OFFER_RATE	STATUS
761	178040	APPROVED BASED ON CREDIT SCORE	21600	5.75	APPROVED
827	123255	APPROVED BASED ON CREDIT SCORE	26400	5.75	APPROVED
512	102975	DENIED BASED ON CREDIT SCORE AND/O...	22000	0	DENIED
608	52110	APPROVED BASED ON CREDIT SCORE	17400	7.5	APPROVED
417	61960	DENIED BASED ON CREDIT SCORE AND/O...	27300	0	DENIED
574	45475	APPROVED BASED ON CREDIT SCORE	31300	7.5	APPROVED
543	52660	APPROVED BASED ON CREDIT SCORE	21700	7.5	APPROVED
608	61495	APPROVED BASED ON CREDIT SCORE	10000	7.5	APPROVED
564	43615	APPROVED BASED ON CREDIT SCORE	21300	7.5	APPROVED
765	61895	DENIED BASED ON CREDIT SCORE AND/O...	65200	0	DENIED
827	61550	APPROVED BASED ON CREDIT SCORE	10800	5.75	APPROVED
635	61870	APPROVED BASED ON CREDIT SCORE	4700	7.5	APPROVED
642	46285	APPROVED BASED ON CREDIT SCORE	19200	7.5	APPROVED

19. Click **x** close the test.

20. Click **x** to close the code file.

Create and Test a Decision

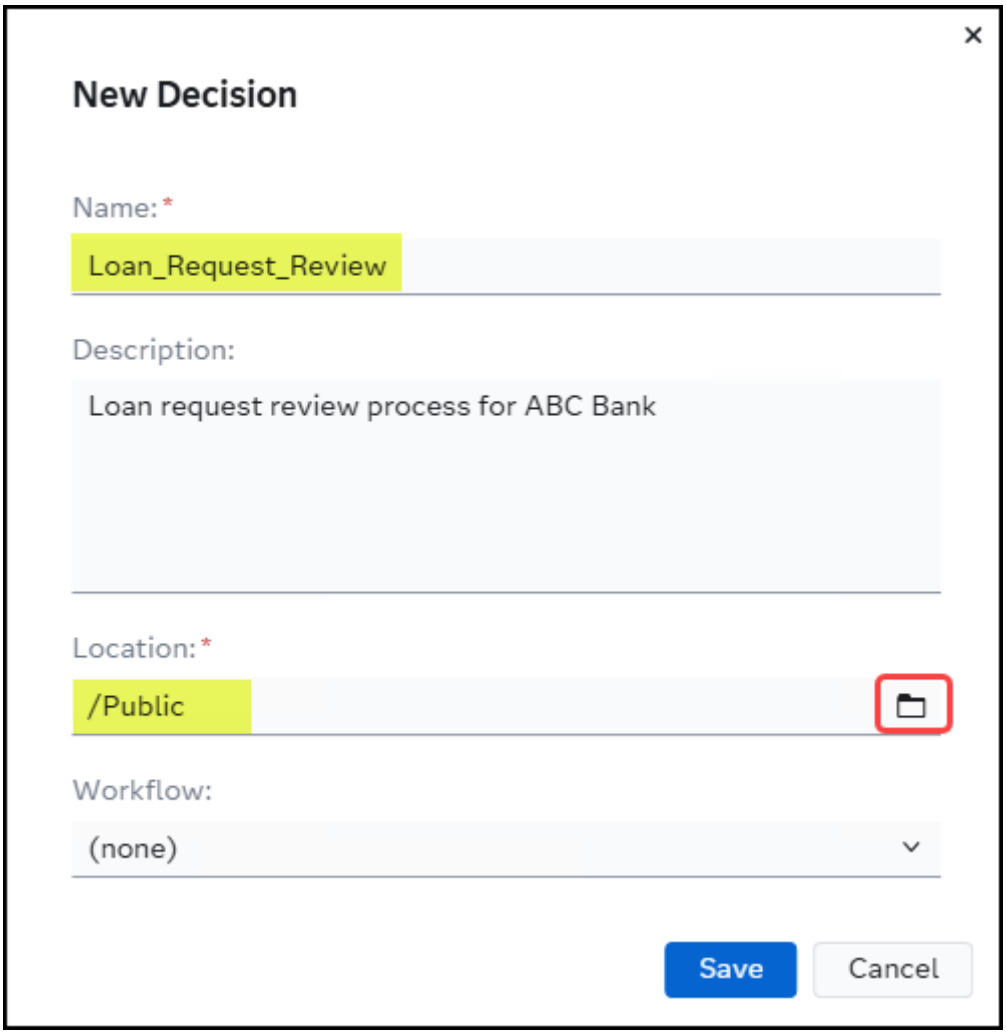
1. Select to open the **Decisions** page.

2. Click **New decision** to create a new decision.

3. Enter the following information for the new decision:

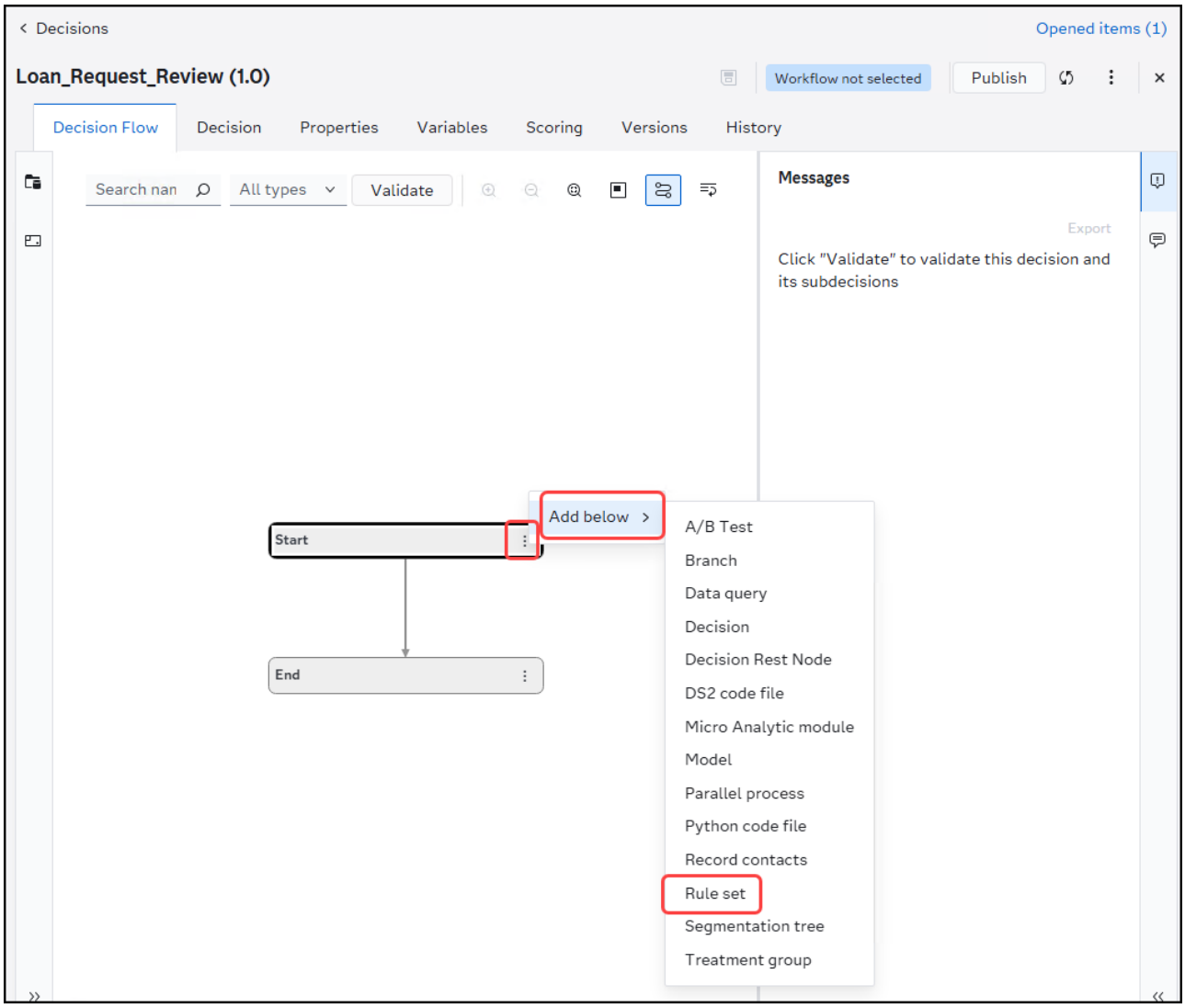
   - Name: **Loan_Request_Review**
   - Description: **Loan request review process for ABC Bank.**
   - Location: **/Public**

   > ☛ Select to navigate to this folder in *SAS Content*.



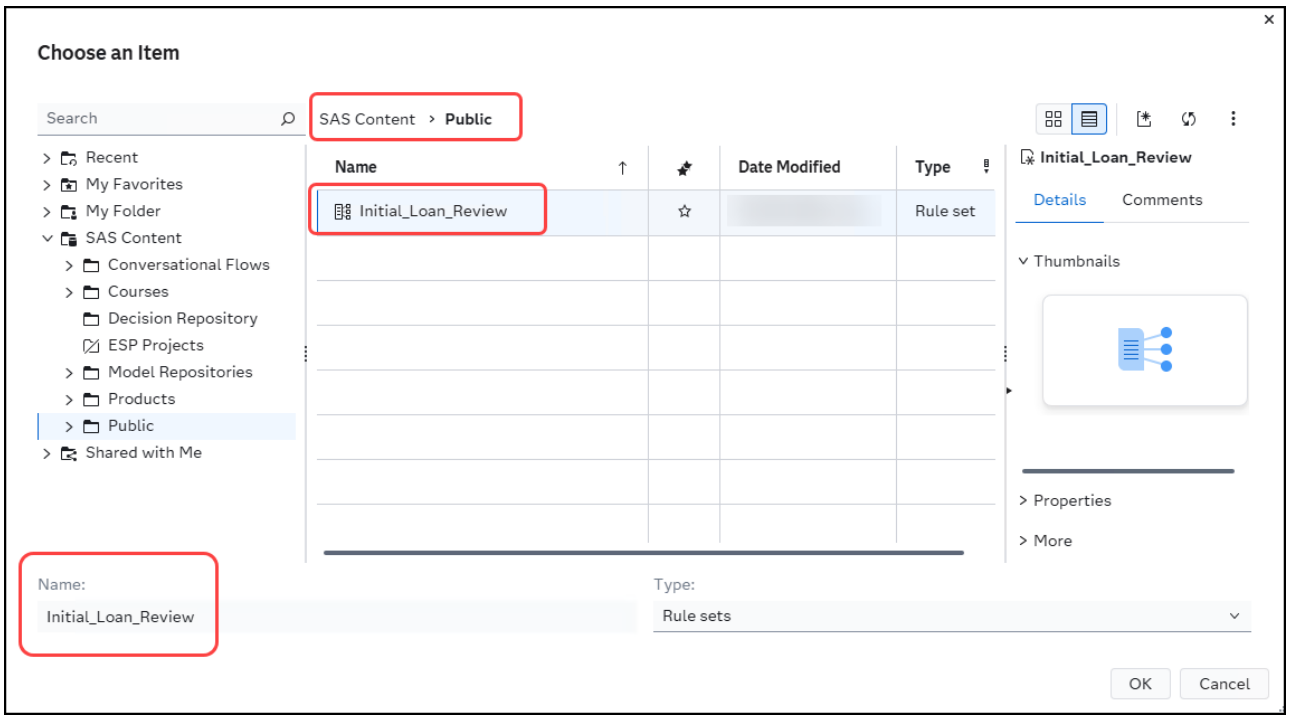
4. Click **Save** to create the decision.

5. On the *Start* node, select ⋮ ➔ **Add below** ➔ **Rule set**.



6. Navigate to the **SAS Content ➔ Public** folder.

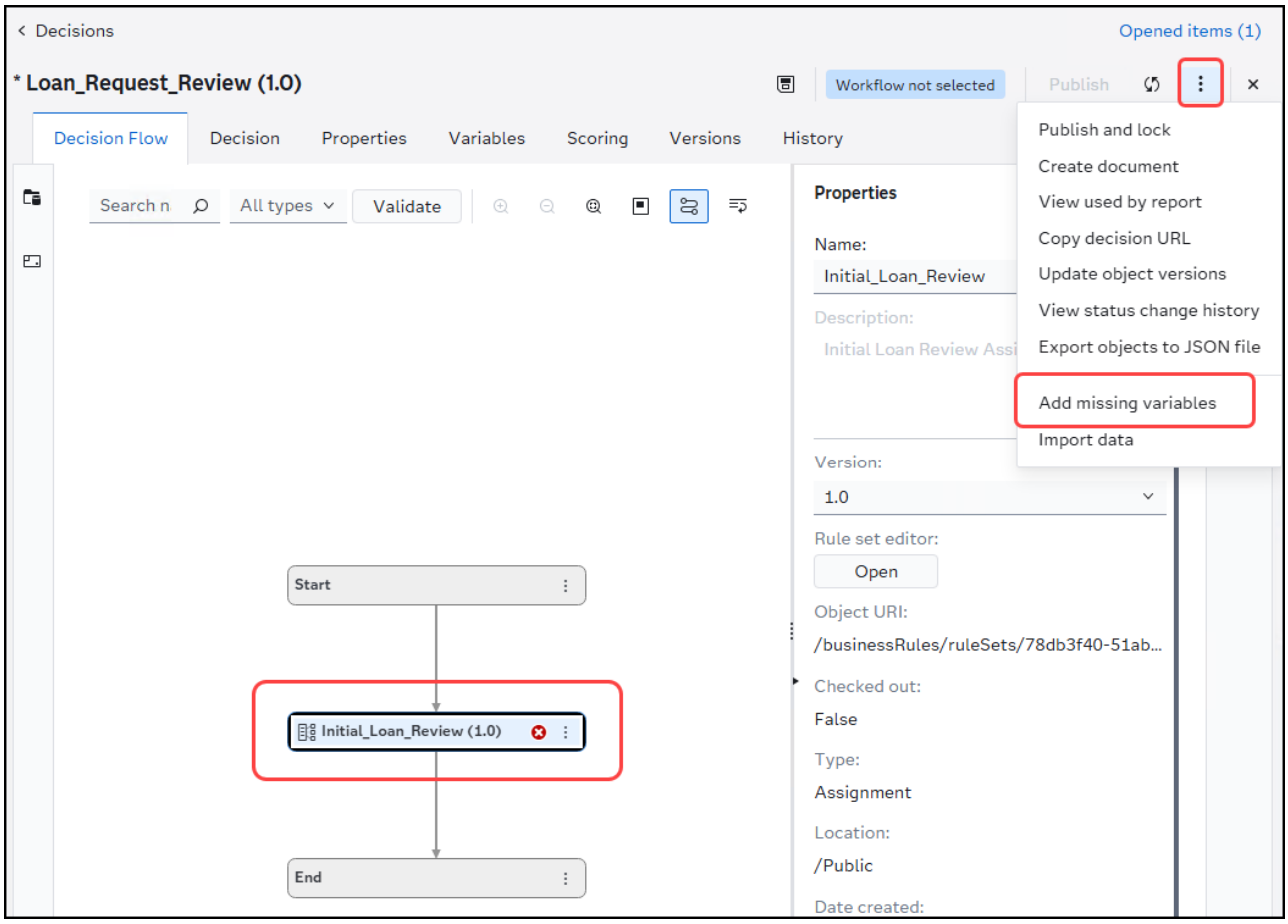
7. Select the **Initial_Loan_Review** rule set.



8. Click **OK** to add the selected rule set to the decision.

> ✎ There is an error since the variables from the rule set are not part of the decision.

9. Select ⋮ ➔ **Add missing variables**.

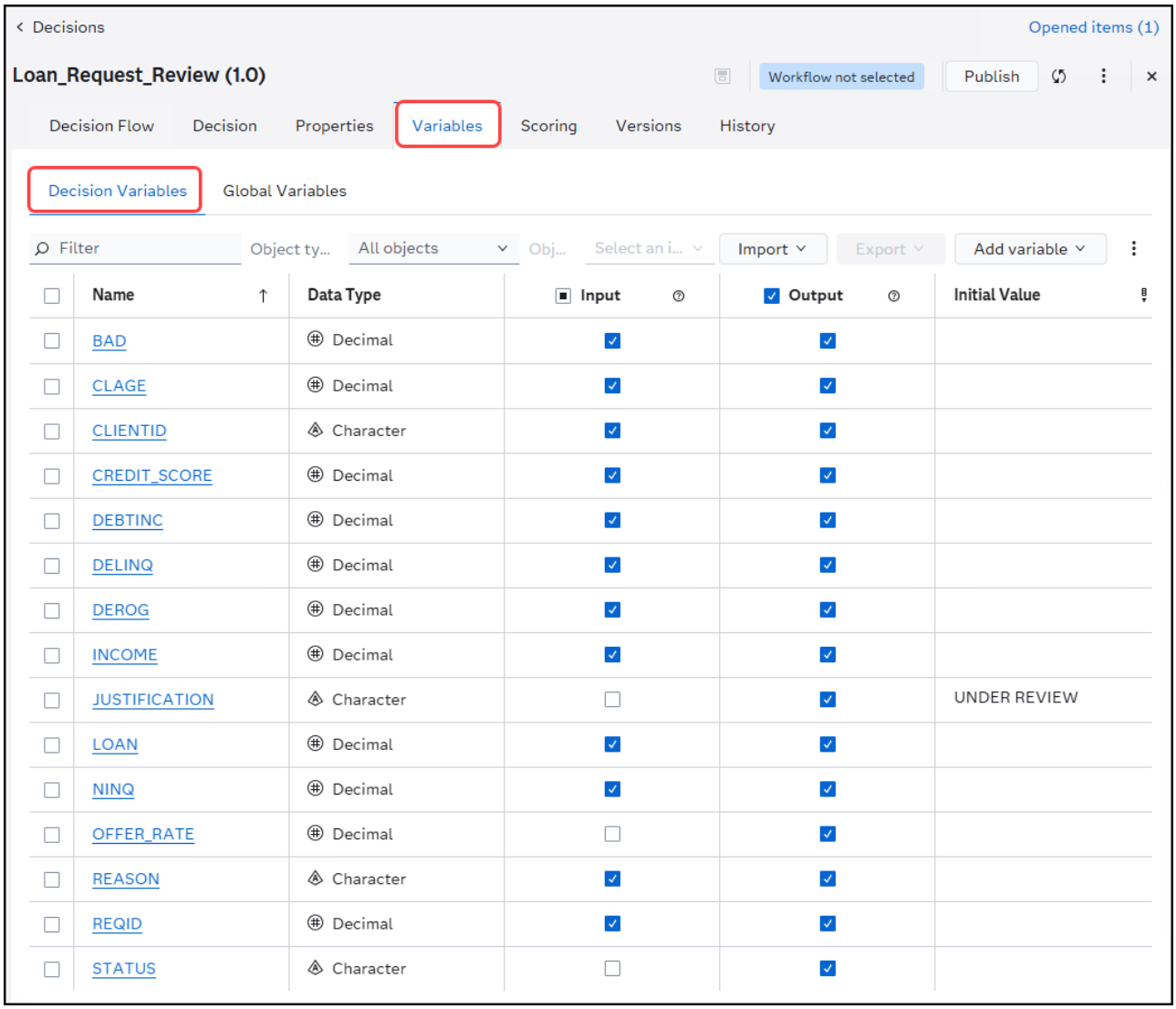


> ✎ There is a setting for *SAS Intelligent Decisioning* where you can select to automatically add missing variables to decision.

10. Click **Add** to add all the missing variables to the decision.

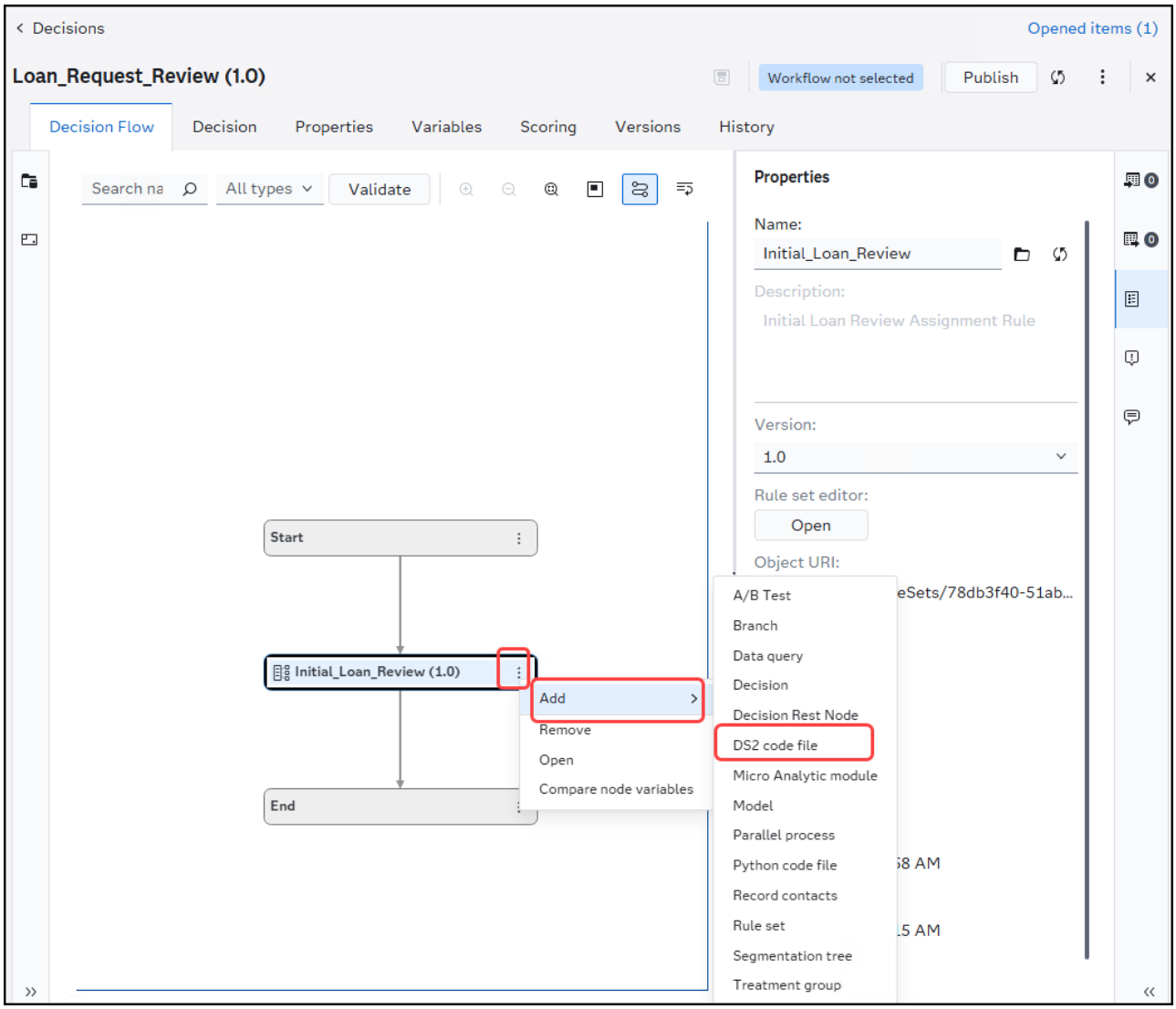
11. Click to save the decision.

12. Select the **Variables** tab to review the added variables.



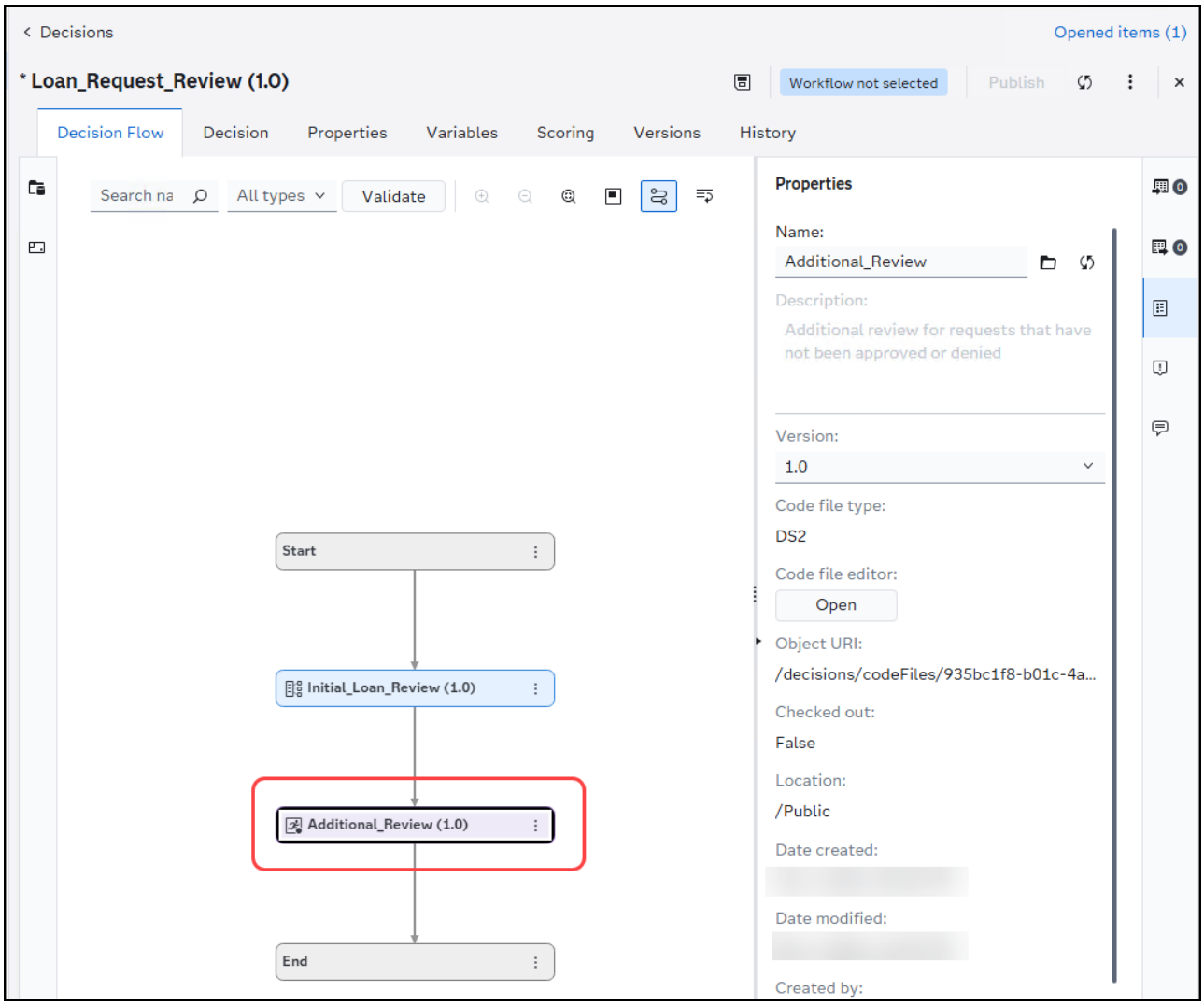
13. Select the **Decision Flow** tab.

14. On the *Initial_Loan_Review* node, select ⋮ → **Add** → **DS2 code file**.



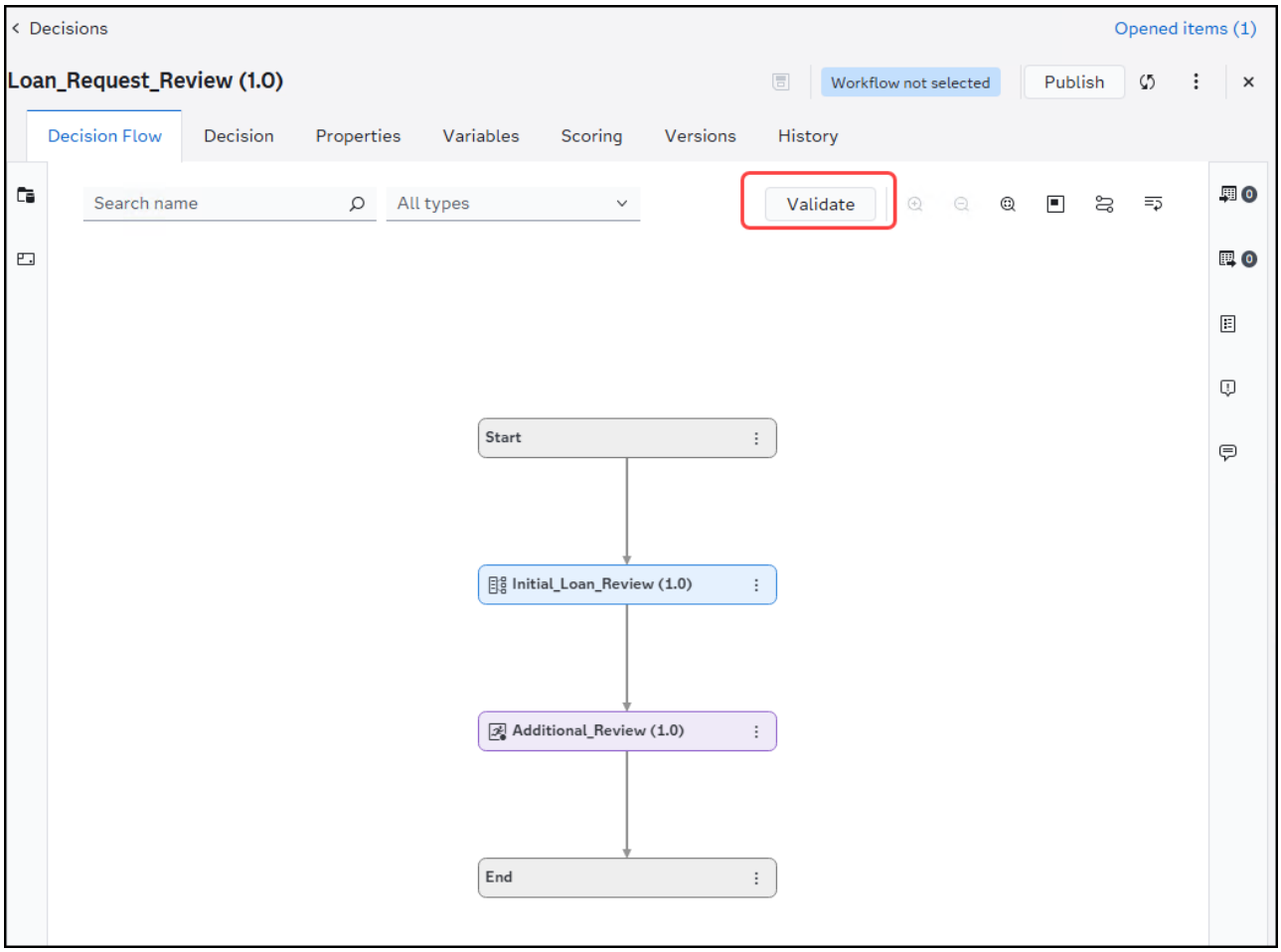
15. Navigate to the **SAS Content ➔ Public** folder.

16. Select the **Additional_Review** DS2 code file and click **OK** to add the selected file to the decision.

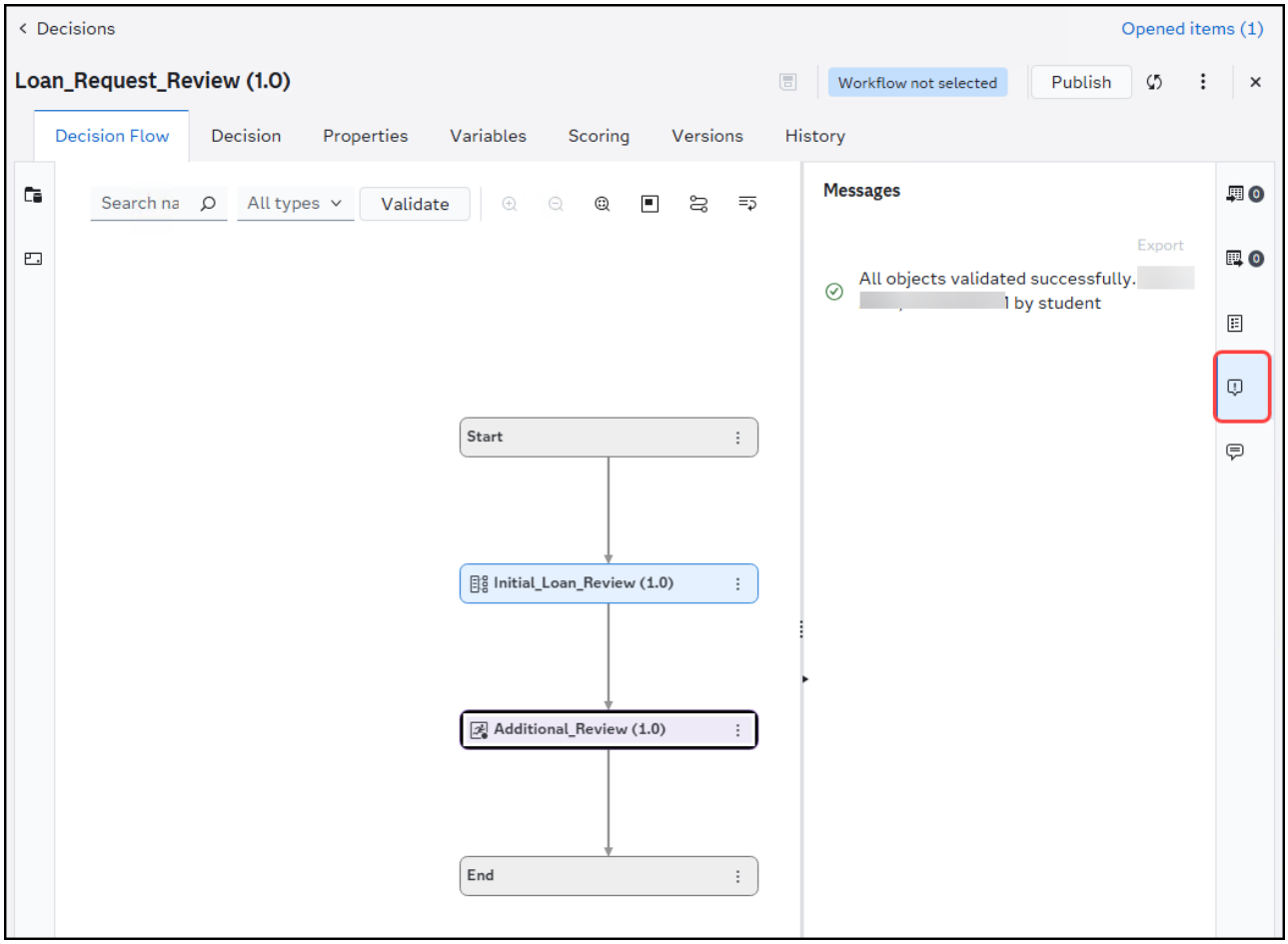


17. Click to save the decision.

18. Click **Validate** to validate the decision.



19. Select on the right pane to confirm if the decision was validated successfully.



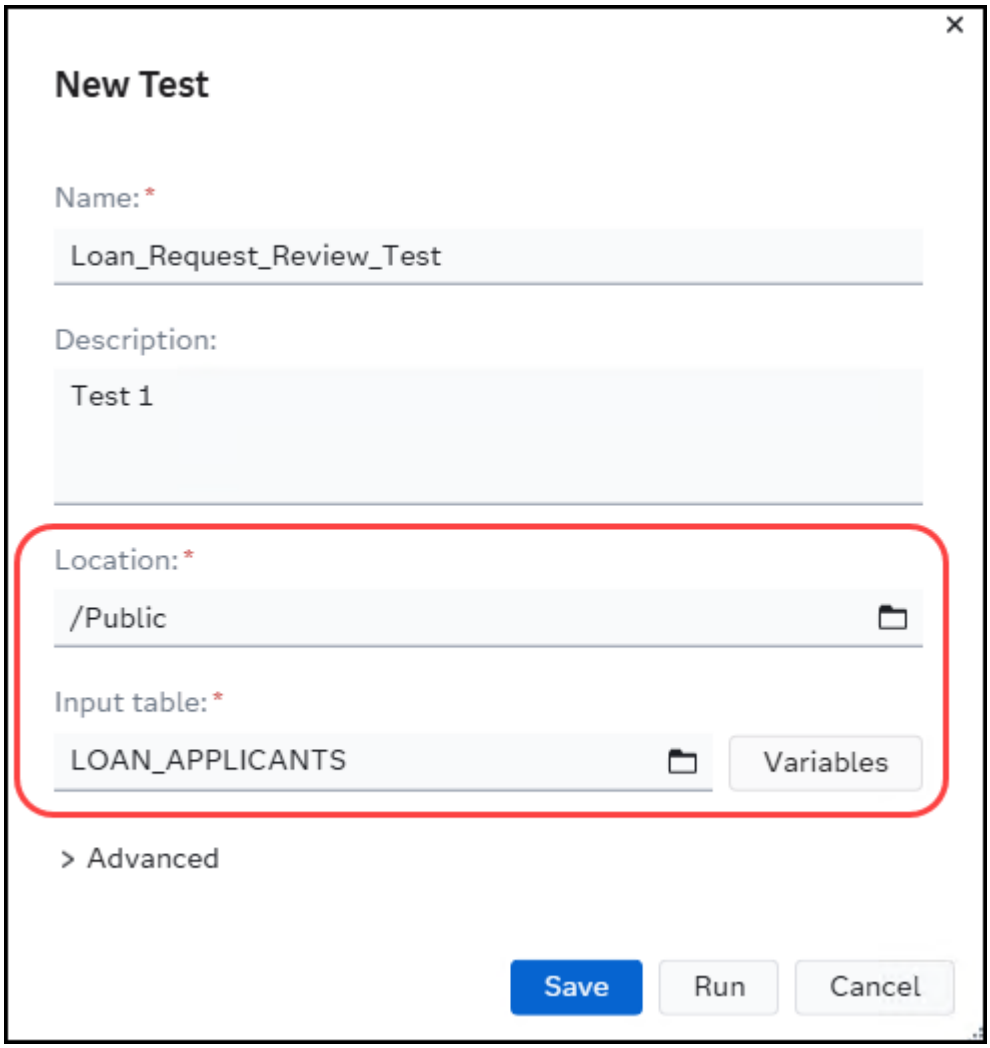
> ✎ The validation step helps identify syntax errors before publishing a decision.

20. To test the decision, select the **Scoring** tab.

21. On the **Tests** sub-tab, click **New Test**.

22. Enter the following information:

    - Name: **Loan_Request_Review_Test**
    - Description: **Test 1**
    - Location: **/Public**
    - Input table: **LOAN_APPLICANTS**.



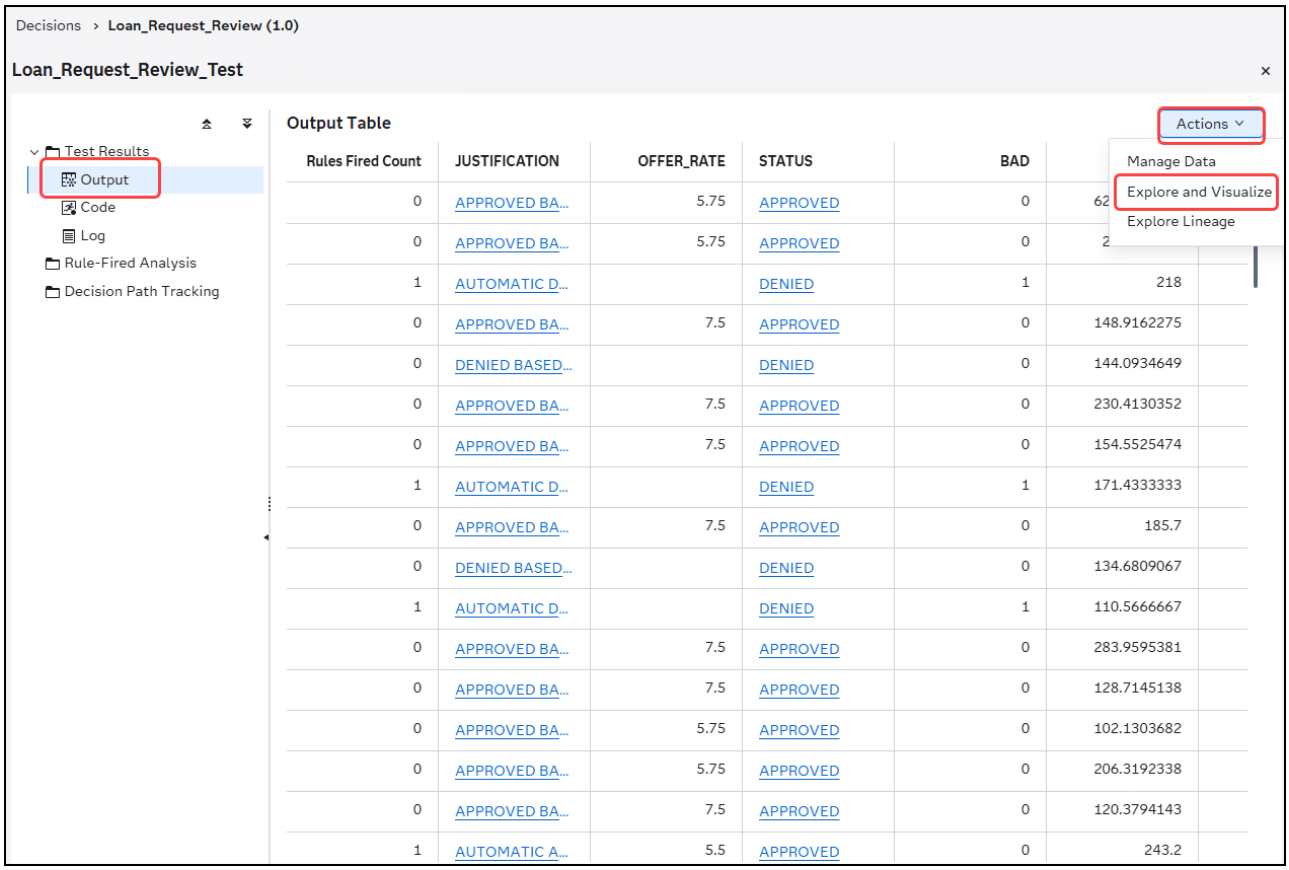
23. Click **Save** to save this test to the decision.

24. Check the newly created test and click **Run** to run the specified test.

25. Once the test run is complete, select to view the test results.

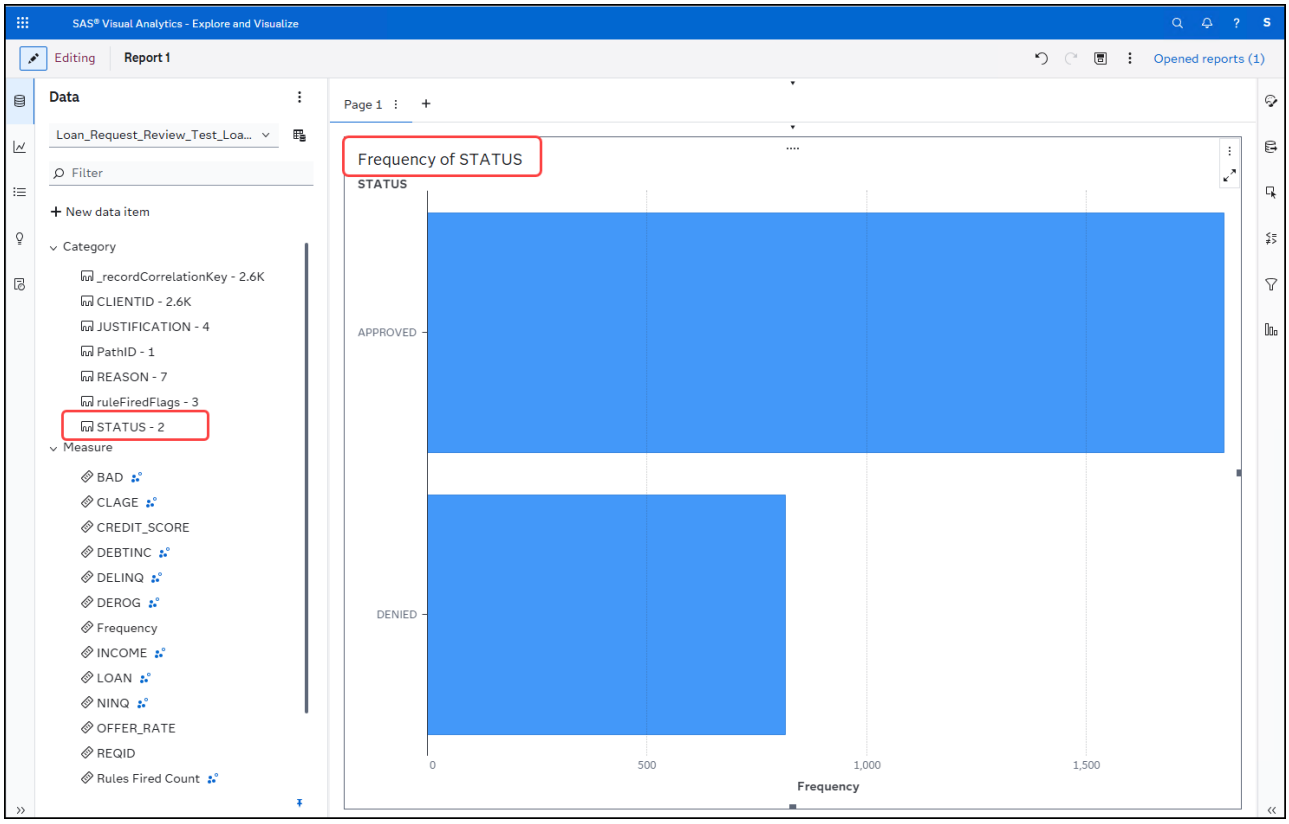
26. Review the results on the **Output** tab.

27. Select **Actions ➔ Explore and Visualize** to open the output table in *SAS Visual Analytics*.



28. Double-click the **STATUS** category to add its values to the report template.

29. Review the resulting bar chart.



> ☛ Select each bar on the report to view the number of records for each status.

30. Select **Opened reports → Close all reports**.

31. Click **Don't save** on the warning dialog.

32. Select ➔ **Build Decisions** to navigate back to *SAS Intelligent Decisioning*.

33. Click **x** close the test.

34. Click **x** to close the decision.

Exercise Completed

You have completed the exercise for **Getting Started with SAS Intelligent Decisioning**.

THANK YOU FOR ATTENDING THIS SESSION!
PLEASE COMPLETE THE EVALUATION!